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HONEST AND ADAPTIVE POINTWISE CONFIDENCE INTERVALS IN LINEAR FUNCTIONAL REGRESSION: AN ANISOTROPIC KERNEL RIDGE REGRESSION APPROACH

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Abstract here.

1. Introduction. We consider the classical model of linear functional regression with data $\{X_i, Y_i\}_{i=1}^n$, where $X_1, \cdots, X_n \sim P_X$ are independently and identically distributed *functions* whose domain is a non-empty compact interval $I \subseteq \mathbb{R}$, and $\{Y_i\}_{i=1}^n$ are scalar response variables, modeled by

$$\mathbb{E}[Y_i|X_i] = \eta_0(X_i) := \gamma_0 + \int_I X_i(t)\beta_0(t)\mathrm{d}t, \tag{1}$$

where $\gamma_0 \in \mathbb{R}$ is the intercept term and $\beta_0 : I \to \mathbb{R}$ is the linear functional, both being unknown parameters. Many well-known methods have been proposed to estimate γ_0 and β_0 from data, such as functional principal component analysis (fPCA) approaches (Ramsay and Silverman, 2005, Chapter 8), (Cai and Hall, 2006) and methods based on reproducing kernel Hilbert spaces (Yuan and Cai, 2010; Cai and Yuan, 2012).

In this paper, we consider the question of constructing *pointwise* confidence intervals on $\eta_0(x)$ which are valuable in quantifying the uncertainty in estimates of the functional model when it is applied to a future data curve x. We study the construction of *honest* confidence intervals, which are functions φ mapping a data curve $x : I \to \mathbb{R}$ to a compact interval $\varphi(x) \subseteq \mathbb{R}$ such that

$$\liminf_{n\to\infty} \inf_{\eta_0\in\Theta} \Pr\left[\eta_0(x) \in \varphi(x)\right] \geq 1-\alpha, \qquad \forall x \in \mathrm{supp}(P_X) \tag{2}$$

where $\alpha \in (0,1)$ is a pre-determined significance level, $\mathrm{supp}(P_X)$ is the support of P_X, Θ is a fixed and known function class and the randomness in Eq. (2) is over $\{(X_i, Y_i)\}_{i=1}^n$. To understand how *adative* an honest confidence interval is, we further study the worst-case mean-squared widths of constructed intervals

$$\sup_{\eta_0\in\Theta} \mathbb{E}_{\eta_0}\left[\int_{\mathrm{supp}(P_X)} \left|\varphi(x)\right|^2 \mathrm{d}P_X(x)\right], \tag{3}$$

where $|\varphi(x)|$ is the Lesbesgue measure of the output interval $\varphi(x)$. We are particularly interested in how the above worst-case risk converges to zero as $n \to \infty$, and how it compares with the mean-squared prediction errors

$$\sup_{\eta_0\in\Theta} \mathbb{E}_{\eta_0}\left[\int_{\mathrm{supp}(P_X)} \int_I [x(t)(\widehat{\eta}_n(t) - \eta_0(t)]^2 \mathrm{d}t\mathrm{d}P_X(x)\right]$$

for classical fPCA or RKHS estimators $\widehat{\eta}$ of η_0.

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1.1. *Our contributions.*

1.2. *Related works.*

1.3. *Notations.* For two sequences of real numbers $\{a_n\}$ and $\{b_n\}$, we write $a_n \lesssim b_n$, or equivalently $a_n = O(b_n)$, if there exists a numerical constant C such that $|a_n| \leq C|b_n|$ for all n. We write $a_n \gtrsim b_n$ or $a_n = \Omega(b_n)$ if $b_n \lesssim a_n$, and $a_n \asymp b_n$ or $a_n = \Theta(b_n)$ if both $a_n \lesssim b_n$ and $a_n \gtrsim b_n$ hold. We denote $L_2(I) := \{x : I \to \mathbb{R} : \int_I x(t)^2 \mathrm{d}t < +\infty\}$ as the class of all square integrable functions on I. For two functions $x, z \in L_2(I)$, $\langle x, z\rangle$ denotes the standard inner product of $L_2(I)$; that is, $\langle x, z\rangle = \int_I x(t)z(t)\mathrm{d}t$. For two real numbers $a, b \in \mathbb{R}$, we write $a \vee b = \max\{a, b\}$ and $a \wedge b = \min\{a, b\}$.

2. Functional regression via kernel ridge regression. In this section, we give an overview of an reproducing kernel Hilbert space (RKHS) approach towards linear functional regression, following the seminal works of Cai and Yuan (2012) which also lays the foundation of methodological and theoretical contributions of this paper. To clarify, all results in this section are relatively well-known in the literature, and we are summarizing them for the purpose of completeness only.

Let $K : I \times I \to \mathbb{R}$ be a real, symmetric, square integrable, and non-negative definite function, known as the *kernel function*. There then exists a reproducing kernel Hilbert space $\mathcal{H} \subseteq L_2(I)$ associated with inner product $\langle \cdot, \cdot\rangle_{\mathcal{H}}$ such that the following reproducing property holds for any $f \in \mathcal{H}$:

$$f(t) = \langle K(t, \cdot), f\rangle_{\mathcal{H}} \qquad \forall t \in I.$$

Assuming $\beta_0 \in \mathcal{H}$. Then a standard kernel ridge regression (KRR) approach to estimate α_0, β_0 from $\{X_i, Y_i\}_{i=1}^n$ is to solve the following penalized optimization problem

$$\widehat{\gamma}_n, \widehat{\beta}_n \in \arg \min_{\gamma\in\mathbb{R},\beta\in\mathcal{H}} \left\{ \frac{1}{n} \sum_{i=1}^{n} (Y_i - \gamma - \langle X_i, \beta\rangle)^2 + \lambda_n \|\beta\|_{\mathcal{H}}^2 \right\}, \tag{4}$$

where $\lambda_n > 0$ is a penalization parameter. Let $\overline{X} = \frac{1}{n}\sum_{i=1}^n X_i$ and $\overline{Y} = \frac{1}{n}\sum_{i=1}^n Y_i$ be the sample averages. It is easy to verify that $\widehat{\gamma}_n = \overline{Y}$ and

$$\widehat{\beta}_n \in \arg \min_{\beta\in\mathcal{H}} \left\{ \frac{1}{n} \sum_{i=1}^{n} (Y_i' - \langle X_i', \beta\rangle)^2 + \lambda_n \|\beta\|_{\mathcal{H}}^2 \right\} \tag{5}$$

where $Y_i' = Y_i - \overline{Y}$ and $X_i' = X_i - \overline{X}$.

2.1. *Reduction to an optimization problem in $L_2(I)$.* The work of Cai and Yuan (2012) makes the observation that Eq. (5) can be equivalently reformulated into an infinite-dimensional optimization problem involving the standard inner product and norm of $L_2(I)$ only, which is helpful for implementation and theoretical analysis purposes. We summarize how such equivalence is built below.

By Mercer theorem, there exists an orthonormal basis $\{\phi_j\}_{j=1}^\infty$ of $L_2(I)$ and a sequence of non-increasing, strictly positive real numbers $\rho_1 \geq \rho_2 \geq \cdots$ such that $\phi_j \in \mathcal{H}$ for all j, and furthermore

$$\langle \phi_j, \phi_k\rangle = \int_I \phi_j(t)\phi_k(t)\mathrm{d}t = \delta_{jk}, \quad \langle \phi_j, \phi_k\rangle_{\mathcal{H}} = \frac{\delta_{jk}}{\rho_j}, \quad \int_I K(t, \cdot)\phi_j(t)\mathrm{d}t = \rho_j\phi_j.$$

where $\delta_{jk} = \mathbf{1}\{j = k\}$ is the Kronecker delta notation. Note that because $\mathcal{H}$ is known, $\{\phi_j, \rho_j\}_{j=1}^{\infty}$ are also known. Following notations in (Cai and Yuan, 2012), we define linear operator $L_{K^{1/2}} : L^2(I) \to L^2(I)$ and its inverse operator $L_{K^{-1/2}}$ as follows:

$$L_{K^{1/2}}(\phi_j) = \sqrt{\rho_j}\phi_j, \quad L_{K^{-1/2}}(\phi_j) = \frac{1}{\sqrt{\rho_j}}\phi_j, \quad \forall j \geq 1.$$

In other words, for any $f = \sum_{j=1}^{\infty} \alpha_j \phi_j \in L^2(I)$ such that $\sum_j \alpha_j^2 < +\infty$, $L_{K^{1/2}}(f) = \sum_{j=1}^{\infty} \alpha_j \sqrt{\rho_j}\phi_j$ and therefore $L_{K^{1/2}} f \in \mathcal{H}$ because $\sum_{j=1}^{\infty} \langle L_{K^{1/2}} f, \phi_j\rangle^2/\rho_j < +\infty$. In fact, (Yuan and Cai, 2010, Proposition 2) shows that for $f \neq 0$ and $L_{K^{1/2}} f \neq 0$, $f \in L^2(I)$ if and only if $L_{K^{1/2}} f \in \mathcal{H}$ and furthermore $\int_I f(t)^2 \mathrm{d}t = \|f\|_2^2 = \|L_{K^{1/2}} f\|_{\mathcal{H}}^2$. The optimal solution $\widehat{\beta}_n$ in Eq. (5) can then be expressed as $\widehat{\beta}_n = L_{K^{1/2}} \widehat{\theta}_n$, where

$$\widehat{\theta}_n \in \arg \min_{\theta \in L_2(I)} \left\{ \frac{1}{n} \sum_{i=1}^{n} (Y_i' - \langle L_{K^{1/2}} X_i', \theta\rangle^2 + \lambda_n \|\theta\|_2^2 \right\}, \tag{6}$$

where $\|\theta\|_2^2 = \int_I \theta(t)^2 \mathrm{d}t$ is the standard squared L_2 norm on I. For brevity, we will also use the notation of $Z_i' = L_{K^{1/2}} X_i'$ throughout this paper.

2.2. *Kernel ridge regression via spectral decomposition.* The infinite-dimensional optimization problem in Eq. (6) can be reduced to a finite-dimensional problem and efficiently solved using Gram matrices, known as the kernel trick. In this section, we show an alternative characterization of $\widehat{\theta}_n$, representing it as a smoothed version of functional PCA which is useful motivating our anisotropic estimators later.

Let

$$T_n := \frac{1}{n} \sum_{i=1}^{n} Z_i' \otimes Z_i'$$

be the sample covariance operator, where $Z_i' = L_{K^{1/2}} X_i'$. By the spectral theorem, there exists orthonormal $\{\widehat{\psi}_j\}_{j=1}^{n}$ in $L_2(I)$ and $\widehat{s}_1 \geq \widehat{s}_2 \geq \cdots \geq \widehat{s}_n \geq 0$ such that

$$T_n = \sum_{j=1}^{n} \widehat{s}_j \widehat{\psi}_j \otimes \widehat{\psi}_j.$$

The estimator $\widehat{\theta}_n$ can then be expressed as

$$\widehat{\theta}_n = (T_n + \lambda_n I)^{-1} \frac{1}{n} \sum_{i=1}^{n} Y_i' Z_i' = \frac{1}{n} \sum_{i=1}^{n} Y_i' \sum_{j=1}^{\infty} \frac{\widehat{\nu}_j(Z_i')}{\widehat{s}_j + \lambda} \widehat{\psi}_j, \tag{7}$$

where $\widehat{\nu}_j(Z_i') = \langle Z_i', \widehat{\psi}_j\rangle = \int_I Z_i'(t) \widehat{\psi}_j(t) \mathrm{d}t$.

3. Assumptions. In this section we list and discuss assumptions that we impose throughout this paper. For some results, additional assumptions are needed: we will remark on them when such additional assumptions are introduced specifically for selected theorems or lemmas.

We first introduce mild regularity conditions to make sure the problem is well-posed.

(A1) Noise variables $\varepsilon_i := Y_i - \eta_0(X_i)$ are independently and identically distributed, centered random variables with $\mathbb{E}[\varepsilon_i^2] = \sigma_\varepsilon^2 < +\infty$ and $\mathbb{E}[|\varepsilon_i|^3] = M_\varepsilon < +\infty$; PD: We need to add the estimation of σ_ε^2, as it enters directly into the construction of the CIs.

(A2) $\beta_0 \in \mathcal{H}$ and $\|\beta_0\|_{\mathcal{H}} \leq 1$, which is equivalent to $\|\theta_0\|_2 = \|L_{K^{-1/2}}\beta_0\|_2 = (\int_I \theta_0(t)^2 \mathrm{d}t)^{1/2} \leq 1$.

(A3) For $x \sim P_X$, $\|L_{K^{1/2}}x\|_2 = (\int_I (L_{K^{1/2}}x)(t))^2 \mathrm{d}t)^{1/2} \leq 1$ almost surely. We also assume that $x \neq 0$ and $L_{K^{1/2}}x \neq 0$ almost surely, to avoid degenerate situations.

We remark that in Assumptions (A2) and (A3) we use one as the upper bounds to $\|\beta_0\|_{\mathcal{H}}$ and $\|x\|_2$ for convenience only. Our constructed honest confidence bands do not require such knowledge and they remain valid with 1 being replaced by any other numerical constants.

We use $\Theta_2(\mathcal{H})$ to denote the class of all regression functionals β_0 that satisfy (A2). Note that $\Theta_2(\mathcal{H})$ depends on the selection of the RKHS $\mathcal{H}$, but not on the covariate distribution P_X.

Let $Z := L_{K^{1/2}}X$ be a random function and P_Z be the distribution of Z, which is induced by both P_X and K. Consider the population covariance operator

$$T := \mathbb{E}_{P_Z}[Z \otimes Z].$$

By the spectral theorem, there exists an orthonormal basis $\{\psi_j\}_{j=1}^{\infty}$ of $L_2(I)$ and $s_1 \geq s_2 \geq \cdots \geq 0$ such that

$$T = \sum_{j=1}^{\infty} s_j \psi_j \otimes \psi_j. \tag{8}$$

Note that, in general, because the population covariance operator T is unknown, $\{\psi_j, s_j\}$ are also unknown.

The following assumptions on imposed directly on the structure of $\{\psi_j\}_j$ and $\{s_j\}_j$.

(B1) There exists numerical constants $r > 1/2$ and $0 < c_s \leq C_s < +\infty$ such that $c_s j^{-2r} \leq s_j \leq C_s j^{-2r}$ for all $j \geq 1$; this also implies that T is invertible;

(B2) There exists a numerical constant $C_p < +\infty$ such that for every $f \in L_2(I)$, $\mathbb{E}[\langle Z, f\rangle^4] \leq C_p(\mathbb{E}[\langle Z, f\rangle^2])^2$;

(B3) There exists numerical constants $\kappa \in [0, 1/2)$ and $C_\nu < +\infty$ such that for every $j \geq 1$, $|\langle Z, \psi_j\rangle| \leq C_\nu n^\kappa \cdot \sqrt{s_j}$ almost surely.

Assumption (B2) is also imposed, in the form of Assumption (b) in (Yuan and Cai, 2010). It is satisfied when X is a Gaussian process.

We remark that, in Assumption (B3), the smaller κ the stronger the assumption is. For the strongest version of $\kappa = 0$, it essentially assumes $|\langle Z, \psi_j\rangle| \lesssim j^{-r}$, which coincides with coefficient decay assumptions adopted in many fPCA analysis. In particular, the results in Cai and Hall (2006) shows that in this case ($\kappa = 0$ in (B3)) fPCA achieves an MSE rate of $n^{-(2r-1)/(2r+1)}$ assuming no decay of $|\langle \theta_0, \psi_j\rangle|$.

In light of Assumption (B3) which is weaker for larger n, we allow the distribution P_X (and therefore the population covariance operator T) to vary for different n, so that more distributions P_X satisfy Assumption (B3) for larger values of n. When necessary, we use the notation P_X^n to emphasize on this fact. We also use P_Z^n to denote the distribution of $Z = L_{K^{1/2}}X$ when $X \sim P_X$.

It should also be remarked that the assumption (B3) with $\kappa = 0$ would greatly simplify the problem, because isotropic weight choices are automatically adaptive. Similar phenomenon revealed in (Liu, Shang and Yang, 2025).

4. Under-smoothed isotropic kernel regression. As a warm-up exercise and a backup method with minimal assumptions, we first consider honest confidence intervals constructed by *under-smoothing* an isotropic kernel ridge estimator. Let $\{\lambda_n\}_{n=1}^{\infty}$ be a sequence of positive penalty parameters that satisfies

$$\lambda_n n \to 0, \;\; \lambda_n n^{1+\delta} \to \infty \quad \text{as } n \to \infty \tag{9}$$

for any $\delta > 0$. Let $\widehat{\gamma}_n = \overline{Y}$ and $\widehat{\beta}_n^{\mathsf{iso}} = L_{K^{1/2}}\widehat{\theta}_n^{\mathsf{iso}}$ where

$$\widehat{\theta}_n^{\mathsf{iso}} = (T_n + \lambda_n I)^{-1}\frac{1}{n}\sum_{i=1}^{n} Y_i' Z_i',$$

with $Y_i' = Y_i - \overline{Y}$, $Z_i' = L_{K^{1/2}}(X_i - \overline{X})$ and $T_n = \frac{1}{n}\sum_{i=1}^{n} Z_i' \otimes Z_i'$. This is an *under-smoothed* kernel ridge regression estimator with $\lambda_n = o(n^{-1})$, while the optimal penalty parameter choice for estimating η_0 in mean-squared errors is $\lambda_n \asymp n^{-2r/(2r+1)}$, per results in Cai and Yuan (2012).

For a test function x in the support P_X^n, an honest and adaptive confidence interval of $\eta(x)$ at significance level $\alpha \in (0,1)$ is constructed as

$$\varphi^{\mathsf{iso}}(x) = [\widehat{\eta}_n^{\mathsf{iso}}(x) \;\pm\; \tau_{\alpha/2}\widehat{\sigma}_n^{\mathsf{iso}}(L_{K^{1/2}}x)]$$

where $\tau_{\alpha/2}$ is the $(1-\alpha/2)$-quantile of the standard Normal distribution, and

$$\widehat{\eta}_n^{\mathsf{iso}}(x) = \widehat{\gamma}_n + \int_I x(t)\widehat{\beta}_n^{\mathsf{iso}}(t)\mathrm{d}t, \qquad \widehat{\sigma}_n(z)^{\mathsf{iso}} = \frac{\sigma_\varepsilon}{\sqrt{n}}\sqrt{\langle z, (T_n + \lambda_n I)^{-1} z\rangle}.$$

4.1. *Honesty of confidence intervals.* For the purpose of our analysis, we also introduce the following quantity $\widetilde{z}_n(z)$ defined as

$$\widetilde{\sigma}_n(z) := \frac{\sigma_\varepsilon}{\sqrt{n}}\sqrt{\langle z, (T_n + \lambda_n I)^{-1} T_n (T_n + \lambda_n I)^{-1} z\rangle},$$

which is smaller than $\widehat{\sigma}_n^{\mathsf{iso}}(z)$ almost surely.

Let $\mathcal{X} := \cup_{n=1}^{\infty} \cap_{m=n}^{\infty} \operatorname{supp}(P_X^m)$ denote the class of all functions which belong to the support of P_X^n for all sufficiently large n.

THEOREM 4.1. *Suppose Assumptions (A1), (A2), (A3), (B1), (B2), (B3) hold, where (B3) holds with $\kappa < \frac{1}{6} - \frac{1}{4r}$. Suppose also that Eq. (9) holds. Then for any $x \in \operatorname{supp}(P_X)$, one can decompose the pointwise prediction error at x as $\widehat{\eta}_n^{\mathsf{iso}}(x) - \eta_0(x) = b_n(x) + v_n(x)$ such that, as $n \to \infty$,*

$$\frac{b_n(x)}{\widehat{\sigma}_n^{\mathsf{iso}}(z)} \;\xrightarrow{a.s.}\; 0, \qquad \frac{v_n(x)}{\widetilde{\sigma}_n(z)} \;\xrightarrow{d}\; \mathcal{N}(0,1),$$

where $z = L_{K^{1/2}}x$. Furthermore, if $\{\varepsilon_i\}_{i=1}^n$ are Normally distributed, then $v_n(x)/\widetilde{\sigma}_n(z)$ is exactly distributed as $\mathcal{N}(0,1)$, and $\sup_{x\in\mathcal{X}} |b_n(x)|/\widehat{\sigma}_n^{\mathsf{iso}}(z) \to 0$ almost surely, and both are true without assuming (B3).

The following corollary is then immediate, noting that $\widetilde{\sigma}_n(z) \le \widehat{\sigma}_n^{\mathsf{iso}}(z)$ almost surely.

COROLLARY 4.1. *Impose conditions in Theorem 4.1. Then*

$$\liminf_{n\to\infty} \inf_{\beta_0 \in \Theta_2(\mathcal{H})} \Pr[\eta_0(x) \in \varphi^{\mathsf{iso}}(x)] \;\ge\; 1 - \alpha, \qquad \forall x \in \mathcal{X}.$$

Furthermore, if $\{\varepsilon_i\}_{i=1}^n$ are Normally distributed, then the following strengthened result holds and it holds without assuming (B3):

$$\liminf_{n\to\infty} \inf_{P_X^n} \inf_{x\in \operatorname{supp}(P_X^n)} \inf_{\beta_0\in\Theta_2(\mathcal{H})} \Pr[\eta_0(x) \in \varphi^{\mathsf{iso}}(x)] \;\ge\; 1 - \alpha.$$

4.2. *Rates of convergence.*

THEOREM 4.2. *Suppose Assumptions (A1)-(A3) and (B1)-(B3) hold, with $\kappa < 1/2$. Then as $n \to \infty$,*

$$\sup_{P_X^n} \sup_{\beta_0 \in \Theta_2(\mathcal{H})} \mathbb{E}_{\eta_0}\left[\int \left|\varphi^{\mathsf{iso}}(x)\right|^2 \mathrm{d}P_X^n(x)\right] \lesssim n^{-\frac{2r-1}{2r}+\delta} \tag{10}$$

for any $\delta > 0$.

4.3. *Minimax optimality.*

THEOREM 4.3. *Fix arbitrary $\alpha \in (0, 1/4)$, $r > 1/2$ and $\kappa = 0$. There exists a distribution P_X satisfying Assumptions (A1), (A3), (B1), (B2), and (B3), such that for any φ satisfying Eq. (2) with respect to the constructed P_X and $\Theta_2(\mathcal{H})$, it holds that*

$$\sup_{\beta_0 \in \Theta_2(\mathcal{H})} \mathbb{E}_{\beta_0}\left[\int \left|\varphi(x)\right|^2 \mathrm{d}P_X(x)\right] \gtrsim n^{-\frac{2r-1}{2r}}.$$

REMARK 4.1. While not directly related, the problem instances constructed in the lower bound proof of Theorem 4.3 would also satisfy the spectral gap assumption (C1) to be introduced in later sections.

We remark that, in the lower bound construction we fix the distribution of X that satisfies all assumptions, including Assumption (B3), with $\kappa = 0$. Because this condition is stronger than Assumption (B3) for any $\kappa > 0$, our lower bound implies lower bounds on relaxed assumptions of $\kappa > 0$ as well. Furthermore, because P_X is fixed, an honest confidence interval can be considered to know P_X in advance: our lower bounds show that even in this case, an $\Omega(n^{-(2r-1)/2r})$ lower bound applies because the honest confidence interval does not know $\beta_0 \in \Theta_2(\mathcal{H})$.

Theorem 4.3 is proved using similar constructions in (Cai and Yuan, 2012), while borrowing ideas from (Cai and Low, 2006) using two-point Le-Cam type arguments using Pinsker's inequality to prove a lower bound on the mean-squared widths of honest confidence intervals that are significantly slower than optimal rates of convergence for functional regression estimators under the mean-squared error criterion. It shows that, while known estimators in the functional regression setting attain $O(n^{-2r/(2r+1)})$ rate for the mean-squared error (Cai and Yuan, 2012), the mean-squared widths of *any* honest confidence interval must scale as $\Omega(n^{-(2r-1)/2r})$, effectively losing one-half degree of smoothness in the spectral decay rate of T.

5. Under-smoothed anisotropic kernel regression. We consider anisotropic extensions to the classical (isotropic) kernel ridge regression, by considering the following estimator

$$\widehat{\theta}_n^{\mathsf{ani}} = (T_n + \Lambda_n)^{\dagger} \frac{1}{n} \sum_{i=1}^{n} Y_i' Z_i'$$

where Λ_n is a positive-semidefinite, self-adjoint covariance operator computed on data that is not necessarily (multiples) of the identity operator, and $C^{\dagger}$ is the Moore-Penrose pseudo-inverse of a not necessarily invertible linear operator C (see, e.g. (Fongi and Gonzalez, 2023) and references therein). Because Λ_n must be entirely calculated from data, we consider the following form of Λ_n, rendering it a finite-rank operator:

$$\Lambda_n = \sum_{j=1}^{n} \lambda_{nj} \widehat{\psi}_j \otimes \widehat{\psi}_j$$

where $\{\lambda_{nj}\}_{j=1}^n \geq 0$ depends only on $\{Z_i'\}_{i=1}^n$ but not Y_i'. With this choice, $\widehat{\theta}_n^{\mathsf{ani}}$ can be written as

$$\widehat{\theta}_n^{\mathsf{ani}} = \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{n} \frac{Y_i' \langle Z_i', \widehat{\psi}_j \rangle}{\widehat{s}_j + \lambda_{nj}} \widehat{\psi}_j.$$

For the purpose of this section, we adopt the choice of

$$\lambda_{nj} = \sqrt{\widehat{s}_j \lambda_n}.$$

With the estimator $\widehat{\eta}_n^{\mathsf{ani}}(\cdot)$ such that $\widehat{\eta}_n^{\mathsf{ani}}(x) = \widehat{\gamma}_n + \int_I (L_{K^{1/2}} x)(t) \widehat{\beta}_n^{\mathsf{ani}}(t) \mathrm{d}t$ where $\widehat{\beta}_n^{\mathsf{ani}} = L_{K^{1/2}} \widehat{\theta}_n^{\mathsf{ani}}$, confidence intervals are constructed as

$$\varphi^{\mathsf{ani}}(x) := [\widehat{\eta}_n^{\mathsf{ani}}(x) \ \pm \ \tau_{\alpha/2} \widehat{\sigma}_n^{\mathsf{ani}}(L_{K^{1/2}} x)]$$

where $\tau_{\alpha/2}$ is the $(1-\alpha/2)$ quantile of the standard Normal distribution, and $\widehat{\sigma}_n^{\mathsf{ani}}(\cdot)$

$$\widehat{\sigma}_n^{\mathsf{ani}}(z) = \frac{\sigma_\varepsilon}{\sqrt{n}} \sqrt{\langle z, (T_n + \Lambda_n)^\dagger T_n (T_n + \Lambda_n)^\dagger z \rangle}.$$

5.1. *Honesty of confidence intervals.*

THEOREM 5.1. *Suppose Assumptions (A1), (A2), (A3), (B1), (B2) hold. Suppose also that λ_n satisfies Eq. (9).If*

$$\frac{\frac{1}{n}\sum_{i=1}^n |g(Z_1, Z_i)|^3}{\sqrt{n}(\frac{1}{n}\sum_{i=1}^n g(Z_1, Z_i)^2)^{3/2}} \ \xrightarrow{p} \ 0 \tag{11}$$

where $g(z, Z_i) = \langle z, (T_n + \Lambda_n)^\dagger Z_i \rangle$ and $Z_i = L_{K^{1/2}} X_i$, then it holds that

$$[\widehat{\sigma}_n^{\mathsf{ani}}(Z_1)]^{-1} (\widehat{\eta}_n^{\mathsf{ani}}(X_1) - \eta_0(X_1)) \ \xrightarrow{d} \ \mathcal{N}(0,1), \tag{12}$$

Furthermore, if $\{\varepsilon_i\}_{i=1}^n$ are Normally distributed, then Eq. (12) holds for uniformly *for all $X_1, \cdots, X_n$ without the need to satisfy Eq. (11).*

COROLLARY 5.1. *Impose the same conditions in Theorem 5.1. Then*

$$\liminf_{n \to \infty} \inf_{\beta_0 \in \Theta_2(\mathcal{H})} \Pr[\eta_0(X_1) \in \varphi^{\mathsf{ani}}(X_1)] \ = \ 1 - \alpha.$$

where X_1 is the first input function in the sample $\{(X_i, Y_i)\}_{i=1}^n$ on which estimation $\widehat{\eta}_n^{\mathsf{ani}}$ and confidence intervals φ^{ani}. Furthermore, if $\{\varepsilon_i\}_{i=1}^n$ are Normally distributed, the above result can be strengthened to

$$\liminf_{n \to \infty} \min_{1 \leq i \leq n} \inf_{\beta_0 \in \Theta_2(\mathcal{H})} \Pr[\eta_0(X_i) \in \varphi^{\mathsf{ani}}(X_i)] \ = \ 1 - \alpha.$$

without the need to satisfy Eq. (11).

REMARK 5.1. In both statements of Theorem 5.1 and Corollary 5.1, we use the notation of X_1 and Z_1 without loss of generality. Because $X_1, \cdots, X_n$ are i.i.d., X_1, Z_1 can be interpreted as an arbitrary input function in the sample $\{X_i, Y_i\}_{i=1}^n$.

5.2. *Rates of convergence.*

THEOREM 5.2. *Impose the same conditions in Theorem 5.1. Then the following holds almost surely as $n \to \infty$:*

$$\frac{1}{n}\sum_{i=1}^{n}\left|\varphi^{\mathsf{ani}}(X_i)\right|^2 \lesssim \sum_{j=1}^{n}\frac{\sqrt{\widehat{s}_j}}{\sqrt{\widehat{s}_j}+\sqrt{\lambda_n}}.$$

where $X_1,\cdots,X_n$ are input functions on which $\widehat{\eta}_n^{\mathsf{ani}}$ and φ^{ani} are estimated. Furthermore, if $\widehat{s}_j \lesssim j^{-2r}$ holds, then the right-hand side of the above inequality is upper bounded by $O(n^{-\frac{2r-1}{2r}+\delta})$ for any $\delta > 0$.

5.3. *Discussion of limitations.* Discuss the limitations and mention truncation is needed.

6. Truncated anisotropic kernel regression. In this section we consider *truncated* anisotropic kernel regression estimators, by setting $\lambda_{nj} \equiv +\infty$ for all $j > J(n)$ for a certain truncation level $J(n) \in [n]$ that depends on n. This effectively mutes all eigenfunctions $\{\widehat{\psi}_j\}_{j>J(n)}$, and the resulting estimator $\widehat{\theta}_n$ admits the following spectral expression

$$\widehat{\theta}_n^{\mathsf{anitr}} = \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{J(n)}\frac{Y_i'\langle Z_i',\widehat{\psi}_j\rangle}{\widehat{s}_j+\lambda_{nj}}\widehat{\psi}_j. \tag{13}$$

For a compressed operator notation, define finite-rank operators T_J and Λ_J as

$$T_J := \sum_{j=1}^{J(n)}\widehat{s}_j\widehat{\psi}_j\otimes\widehat{\psi}_j, \qquad \Lambda_J := \sum_{j=1}^{J(n)}\lambda_{nj}\widehat{\psi}_j\otimes\widehat{\psi}_j.$$

Then $\widehat{\theta}_n^{\mathsf{anitr}}$ can be represented as

$$\widehat{\theta}_n^{\mathsf{anitr}} = (T_J+\Lambda_J)^{\dagger}\frac{1}{n}\sum_{i=1}^{n}Y_i'Z_i',$$

where $C^{\dagger}$ is the Moore-Penrose pseudo-inverse of a not necessarily invertible linear operator C.

Following the previous section, one option for the choice of the anisotropic Ridge penalty parameters is $\lambda_{nj} = \sqrt{\widehat{s}_j\lambda_n}$ with λ_n satisfying Eq. (9), and truncate at the level of $J(n) \le n$. Let $\{\omega_n\}_{n=1}^{\infty}$ be a sequence of slowly increasing positive constants such that

$$\omega_n \to \infty, \qquad \omega_n n^{-\delta} \to 0 \tag{14}$$

as $n \to \infty$, for any $\delta > 0$. The estimator is then $\widehat{\eta}_n^{\mathsf{anitr}}(\cdot) = \widehat{\gamma}_n + \langle\cdot,\widehat{\beta}_n^{\mathsf{anitr}}\rangle$ where $\widehat{\beta}_n^{\mathsf{anitr}} = L_{K^{1/2}}\widehat{\theta}_n^{\mathsf{anitr}}$, and $\langle f,g\rangle = \int_I f(t)g(t)\mathrm{d}t$. Afterwards, for a test function $x \in \mathrm{supp}(P_X)$, a confidence interval for $\eta_0(x)$ is constructed as

$$\varphi^{\mathsf{anitr}}(x) = [\widehat{\eta}_n^{\mathsf{anitr}}(x) \,\pm\, \tau_\alpha\widehat{\sigma}_n^{\mathsf{anitr}}(L_{K^{1/2}}x) + \omega_n\|\widehat{\Pi}_J^{\perp}(L_{K^{1/2}}x)\|_2],$$

where τ_α is the $(1-\alpha/2)$ quantile of the standard Normal distribution, $\widehat{\Pi}_J^{\perp}z = (I - \sum_{j=1}^{J(n)}\widehat{\psi}_j\otimes\widehat{\psi}_j)z$, and $\widehat{\sigma}_n^{\mathsf{anitr}}$ is defined as

$$\widehat{\sigma}_n^{\mathsf{anitr}}(z) = \frac{\sigma_\varepsilon}{\sqrt{n}}\sqrt{\langle z,(T_J+\Lambda_J)^{\dagger}T_J(T_J+\Lambda_J)^{\dagger}z\rangle}$$

6.1. *Spectrum stability of the sample covariance operator.* In this section, we establish rigorously the connections between the (leading) spectrum of T and T_n, with cutoff thresholds set at levels that can be as deep as $o(\sqrt{n})$. Such discussion is important for the analysis of truncated anisotropic kernel regression estimators, for two primary reasons:

1. To establish asymptotic Normality for the widest class of testing functions $x \in \mathrm{supp}(P_X)$ possible, we must use the central limit theorem for independent but not identically distributed random variables and verify classical conditions such as the Lyapnov condition or the Lindeberg condition. This involves understanding the distributions of "influence functions" in the variance term, which must be connected to their population variants in order to avoid statistical correlation between Z_i and T_n;
2. The condition (D1) to be introduced in the next section, unlike the condition (A2) that $\int_I \theta_0(t)^2 \mathrm{d}t \le 1$, is *not* basis invariant. This means that on a different orthonormal basis, such as the eigenfunctions $\{\widehat{\psi}_j\}_{j=1}^n$ of the sample covariance operator T_n, the sum $\sum_{j\ge 1} |\langle Z_j, \widehat{\psi}_j\rangle|$ may not be bounded by B, which leads to sub-optimal convergence rates if such sums indeed scale polynomially with n. Therefore, we must rigorously compare the spectrum $\{\psi_j\}$ and $\{\widehat{\psi}_j\}$ of the population and sample covariance operators, to ensure that $\sum_{j\ge 1} |\langle Z_j, \widehat{\psi}_j\rangle|$ remains bounded so that the bias of kernel Ridge regression can be effectively controlled.

To establish such spectrum stability, we need the following assumption which lower bounds spectral gaps between consecutive eigenvalues on the population covariance operator:

(C1) There exists a numerical constant $c_g > 0$ such that for all $j \ge 1$ PD: need the gap condition for all $1 \le j \le \lceil C_J J(n) \rceil$, that is slightly above the truncation index, $s_{j+1} - s_j \ge c_g j^{-2r-1}$, where $\{s_j\}_{j=1}^\infty$ are eigenvalues (listed in descending order) of the population covariance operator $T = \mathbb{E}_{P_X}[L_{K^{1/2}} X \otimes L_{K^{1/2}} X]$.

We further impose the following condition on the estimator's choice of truncation levels.

(C2) There exists a constant $q \in (0, 1/2)$ and $0 < c_q \le C_q < +\infty$ such that $c_q n^q \le J(n) \le C_q n^q$ for all $n \ge 3$.

(C3) **(Anti-concentration)**. There exists a constant $C_\epsilon > 0$ and a small constant $\lambda_\epsilon > 0$ such that, for any $\lambda \in (0, \lambda_\epsilon]$ and $J \in \mathbb{N}$, it holds that

$$\Pr_{Z\sim\mathcal{P}_Z}\left[\frac{\sum_{j=1}^J a_j(\lambda)\langle Z, \psi_j\rangle^2}{\sum_{j=1}^J a_j(\lambda) s_j} \le \epsilon\right] \le C_\epsilon \epsilon,$$

where $a_j(\lambda) := s_j/(s_j+\lambda)$ is a non-negative, fixed quantity depending only on λ. Note that, here the denominator is the expected value of the numerator.

Recall that the population covariance operator $T = \mathbb{E}[Z \otimes Z]$ and the sample covariance operator $T_n = \frac{1}{n}\sum_{i=1}^n Z_i \otimes Z_i$ admit spectral decomposition of $T = \sum_{j=1}^\infty s_j \psi_j \otimes \psi_j$ and $T_n = \sum_{j=1}^n \widehat{s}_j \widehat{\psi}_j \otimes \widehat{\psi}_j$, where $\{\psi_j\}_{j=1}^\infty$ is an orthonormal basis of $L_2(I)$ and $\{\widehat{\psi}_j\}_{j=1}^n$ are orthonormal in $L_2(I)$.

LEMMA 6.1. *Suppose Assumptions (A3), (B1), (B2), (B3), (C1) and (C2) hold, with parameters $\kappa < 1/4$ and $q \in (0, 1/2)$. Fix an arbitrary small constant $c \ge 2$. Then there exist constants $C_J, L_J < +\infty$ depending only on c and other parameters in assumptions, such that for every $n \ge 3$, with probability $1 - n^{-c}$ the following holds: there exists an injection*

$\varpi : [\lceil C_J J(n)\rceil] \to \mathbb{N}$ *which is a surjection on* $[J(n)]$, [1] *such that for every* $j \in [\lceil C_J n\rceil]$, *the following hold:*

1. $j/C_J \le \varpi(j) \le C_J j$;
2. $|\widehat{s}_{\varpi(j)}/s_j - 1| \le L_J j \log^3 n/\sqrt{n}$;
3. *For every* $k \in \mathbb{N}$, $k \neq j$, $|\langle \widehat{\psi}_{\varpi(j)}, \psi_k\rangle| \le L_J \Delta_\psi(j,k)$, *where*

$$\Delta_\psi(j,k) = \frac{\log n \log(3j)\log(3k)}{\sqrt{n}} \times \begin{cases} k^r/j^r, & \text{if } k < j/C_J; \\ j^r/k^r, & \text{if } k > C_J j; \\ \sqrt{jk}/|j-k|; & \text{if } j/C_J \le k \le C_J j \text{ and } j \neq k. \end{cases}$$

Furthermore, if $q < 1/4$ *then* ϖ *can be chosen as the bijection* $\varpi(j) = j$ *for all* $j \le \lceil C_J J(n)\rceil$.

Mention why this can't be done using Davis Kahan, citing existing fPCA papers.

Mention also that the $q < 1/4$ bijection result is for information only and *not* necessary for our results.

The following lemma, as a consequence of Lemma 6.1 and Assumptions (B2) and (B3) regulating the behaviors of $\langle Z, \psi_j\rangle$, upper bounds the discrepancy between coefficients on $\{\psi_j\}$ and $\{\widehat{\psi}_j\}$ of a test function $z = L_{K^{1/2}} x$.

LEMMA 6.2. *Impose the same set of conditions in Lemma 6.1. Let* $J = J(n) \asymp n^q$ *and* $\underline{J} = J(n)/C_J$. *For each* $M \in \{2,4\}$, *conditioned on results in Lemma 6.1, it holds that*

$$\sup_{\ell \le \underline{J}} \sup_{z \in \mathrm{supp}(P_Z^n)} \left|\langle z, \widehat{\psi}_\ell\rangle^M - \langle z, \psi_j\rangle^M\right| \lesssim n^{M\kappa} \times \frac{J^{1-rM}\log^4 n}{\sqrt{n}},$$

where j *is the unique integer satisfying* $\varpi(j) = \ell$, *whose existence is guaranteed by Lemma 6.1. Furthermore,*

$$\sup_{\ell \le \underline{J}} \left| \mathbb{E}_{Z\sim P_Z^n}\left[\langle Z, \widehat{\psi}_\ell\rangle^M\right] - \mathbb{E}_{Z\sim P_Z^n}\left[\langle Z, \psi_j\rangle^M\right]\right| \lesssim \frac{J^{1-rM}\log^4 n}{\sqrt{n}}.$$

Note that we only need $M = 2$. But keeping $M = 4$ which might be interesting for future research.

6.2. *Honesty of confidence intervals.* The following theorem establishes the limiting distribution of $\widehat{\eta}_n^{\mathsf{anitr}}$.

THEOREM 6.1. *Impose all assumptions and conditions in Lemma 6.1. Suppose* Λ_n *is constructed as* $\lambda_{nj} = \sqrt{\widehat{s}_j \lambda_n}$ *for all* $j \le J(n)$, *where* λ_n *satisfies Eq. (9). Suppose also that Assumptions (A1), (A2) hold, and* $\kappa < (1-q)/2$. *Then*

$$\lim_{n\to\infty} \sup_{P_X^n} \sup_{x \in \mathrm{supp}(P_X^n)} \sup_{\beta_0 \in \Theta_2(\mathcal{H})} \sup_{t \in \mathbb{R}} \left| \Pr\left[\frac{\widehat{\eta}_n^{\mathsf{anitr}}(x) - \eta_0(x) + \langle \widehat{\Pi}_J^\perp z, \theta_0\rangle}{\widehat{\sigma}_n^{\mathsf{anitr}}(z)} \le t\right] - \Phi(t)\right| = 0,$$

where $z = L_{K^{1/2}} x$, $\theta_0 = L_{K^{-1/2}}\beta_0$, $\widehat{\Pi}_J^\perp = I - \sum_{j=1}^{J(n)} \widehat{\psi}_j \otimes \widehat{\psi}_j$, *and* $\Phi(\cdot)$ *is the CDF of the standard Normal distribution. The probability is over the randomness of both* $\{X_i\}_{i=1}^n$ *and* $\{\varepsilon_i\}_{i=1}^n$.

[1] This means that 1) for every $j, j' \in [\lceil C_J J(n)\rceil]$, $j \neq j'$ implies $\varpi(j) \neq \varpi(j')$, and 2) for every $\ell \in [J(n)]$, there exists $j \in [\lceil C_J J(n)\rceil]$ such that $\varpi(j) = \ell$.

Furthermore, if $\frac{1}{2r} < q < \frac{1}{2}$ *and Assumption (C3) holds, then for every* $\epsilon > 0$*, there exist measurable sets* $\mathcal{X}_1 \subseteq \text{supp}(P_X^1), \mathcal{X}_2 \subseteq \text{supp}(P_X^2), \cdots$ *satisfying* $P_X^1(\mathcal{X}_1), P_X^2(\mathcal{X}_2), \cdots \geq 1 - \epsilon$*, such that*

$$\lim_{n\to\infty} \sup_{P_X^n} \sup_{x\in\mathcal{X}_n} \sup_{\beta_0\in\Theta_2(\mathcal{H})} \sup_{t\in\mathbb{R}} \left| \Pr\left[\frac{\widehat{\eta}_n^{\mathsf{anitr}}(x) - \eta_0(x)}{\widehat{\sigma}_n^{\mathsf{anitr}}(z)} \leq t \right] - \Phi(t) \right| = 0.$$

REMARK 6.1. Taking q to be close to $1/2$, the condition $1/2r < q < 1/2$ amounts to $r > 1$.

Note that

$$\left|t_n(z)\right| = \left|\langle z, \widehat{\Pi}_J^{\perp}\theta_0\rangle\right| \leq \|\widehat{\Pi}_J^{\perp} z\|_2 \cdot \|\theta_0\|_2 \leq \omega_n \|\widehat{\Pi}_J^{\perp} z\|_2, \tag{15}$$

where the first inequality holds from Cauchy-Schwarz inequality and the fact that $\widehat{\Pi}_J^{\perp}$ is a self-adjoint, projection operator, and the last inequality holds for sufficiently large n because $\|\theta_0\|_2 \leq 1$ and $\omega_n \to \infty$ as $n \to \infty$.

COROLLARY 6.1. *Impose the same set of assumptions and conditions in Theorem 6.1. Then*

$$\liminf_{n\to\infty} \inf_{P_X^n} \inf_{x\in\text{supp}(P_X^n)} \inf_{\beta_0\in\Theta_2(\mathcal{H})} \Pr[\eta_0(x) \in \varphi^{\mathsf{anitr}}(x)] \;\geq\; 1 - \alpha.$$

Furthermore, if $\frac{1}{2r} < q < \frac{1}{2}$*, then the following holds for the choice of* $\omega_n \equiv 0$*: for any* $\epsilon > 0$*, there exist measurable sets* $\mathcal{X}_1 \subseteq \text{supp}(P_X^1), \mathcal{X}_2 \subseteq \text{supp}(P_X^2), \cdots$ *satisfying* $P_X^1(\mathcal{X}_1), P_X^2(\mathcal{X}_2), \cdots \geq 1 - \epsilon$*, such that for any sequence* $x_1 \in \mathcal{X}_1, x_2 \in \mathcal{X}_2, \cdots$*,*

$$\lim_{n\to\infty} \inf_{\beta_0\in\Theta_2(\mathcal{H})} \Pr[\eta_0(x) \in \varphi^{\mathsf{anitr}}(x)] \;=\; 1 - \alpha.$$

6.3. *Rates of convergence.*

THEOREM 6.2. *Impose all assumptions and conditions in Lemma 6.1. Suppose* Λ_n *is constructed as* $\lambda_{nj} = \sqrt{\widehat{s}_j \lambda_n}$ *for all* $j \leq J(n)$*, where* λ_n *satisfies Eq. (9). Then as* $n \to \infty$*,*

$$\sup_{P_X^n} \sup_{\beta_0\in\Theta_2(\mathcal{H})} \mathbb{E}\left[\int \left|\varphi^{\mathsf{anitr}}(x)\right|^2 \mathrm{d}P_X^n(x) \right] \lesssim n^{-\frac{2r-1}{2r}+\delta} + n^{2q-\frac{3}{2}+\delta} + n^{-q(2r-1)+\delta} + n^{1-c}.$$

for any $\delta > 0$*, where* $c \geq 2$ *is the parameter in the failure probability in Lemma 6.1.*

REMARK 6.2. If $\frac{1}{2r} < q < \frac{1}{4r} + \frac{1}{4}$ then the right-hand side of the above inequality is asymptotically on the order of $O(n^{-(2r-1)/2r+\delta})$ for any $\delta > 0$.

7. More adaptivity in anisotropic regularization weights.

(D1) There exists a numerical constant $B_1 < +\infty$ such that $\sum_{j=1}^{\infty} |\langle\theta_0, \psi_j\rangle| \leq B_1$, where $\{\psi_j\}_{j=1}^{\infty}$ are eigenfunctions of the population covariance operator T, and $\theta_0 = L_{K^{-1/2}}\beta_0$ for some $\beta_0 \in \mathcal{H}$.

We use $\Theta_1(\mathcal{H}) \subseteq \Theta_2(\mathcal{H})$ to denote all unknown linear functionals in $\Theta_2(\mathcal{H})$ that also satisfies Assumption (D1), for a pre-determined constant upper bound B_1. This assumption is also implicitly imposed, albeit in a different form, in the works of Liu, Shang and Yang (2025) to construct pointwise confidence intervals for classical kernel estimators.

Maybe some remarks on Fourier Basis, or alignment arising from Gaussian processes.

Let $\mu_n > 0$ be a sequence of penalty parameters that decay slightly slower than λ_n, satisfying

$$\mu_n n^{2r/(2r+1)} \to 0, \qquad \mu_n n^{2r/(2r+1)+\delta} \to \infty \tag{16}$$

for all $\delta > 0$. We consider $\widehat{\theta}_n^{\mathsf{adptr}}$ as solution to Eq. (13), with $\Lambda_n = \sum_{j=1}^{J(n)} \lambda_{nj}(z)\widehat{\psi}_j \otimes \widehat{\psi}_j$ being defined as

$$\lambda_{nj}(z) = \begin{cases} \widehat{s}_j\sqrt{\mu_n}/|\langle z, \widehat{\psi}_j\rangle|, & \text{if } \langle z, \widehat{\psi}_j\rangle^2 \geq 2\widehat{s}_j; \\ \sqrt{\widehat{s}_j\mu_n}, & \text{otherwise;} \end{cases} \tag{17}$$

and truncation occurs at $J(n) \leq n$. Note that in this estimator, the choice of Λ_n and the penalization parameters λ_{nj} are *adaptive* to the testing function $z = L_{K^{1/2}}x$, which is different from the previous estimators that do not depend on z.

We remark that the constant 2 in the first prong of the definition of Eq. (17) can be changed to any constant without affecting the analysis of this estimator. This also implies that, if Assumption (B3) holds with $\kappa = 0$, then there is no need to do have anisotropic regularization parameters depending on the test function z, and one can instead set simply $\lambda_{nj} = \mu_n \wedge \sqrt{\widehat{s}_j\mu_n}$.

For a testing function z and point estimate $\widehat{\eta}_n^{\mathsf{adptr}}(x) = \widehat{\gamma}_n + \langle L_{K^{1/2}}z, \widehat{\theta}_n^{\mathsf{adptr}}\rangle$, a confidence interval of $\eta_0(x)$ is constructed as

$$\varphi^{\mathsf{adptr}}(x) = [\widehat{\eta}_n^{\mathsf{adptr}}(x) \ \pm \ \tau_\alpha\widehat{\sigma}_n^{\mathsf{adptr}}(L_{K^{1/2}}x) + \omega_n(\sqrt{\mu_n} + \|\widehat{\Pi}_J^{\perp}(L_{K^{1/2}}x)\|_2)],$$

where τ_α is the $(1-\alpha/2)$ quantile of the standard Normal distribution, and $\widehat{\sigma}_n^{\mathsf{adptr}}$ is defined as

$$\widehat{\sigma}_n^{\mathsf{adptr}}(z) = \frac{\sigma_\varepsilon}{\sqrt{n}}\sqrt{\langle z, (T_J + \Lambda_J)^\dagger T_J (T_J + \Lambda_J) z\rangle} = \frac{\sigma_\varepsilon}{\sqrt{n}}\sqrt{\sum_{j=1}^{J(n)} \frac{\langle z, \widehat{\psi}_j\rangle^2 \widehat{s}_j}{(\widehat{s}_j + \lambda_{nj})^2}}.$$

7.1. *Honesty of confidence intervals.*

THEOREM 7.1. *Impose all assumptions and conditions in Lemma 6.1. Suppose Λ_n is constructed in Eq. (17) for all $j \leq J(n)$, where μ_n satisfies Eq. (16) and q in Assumption (C2) satisfies $q < 1/2$. Suppose also that Assumptions (A1), (A2), (D1) hold, and $\kappa < (1-q)/2$. Then $\widehat{\eta}_n^{\mathsf{adptr}}(x) - \eta_0(x) + \langle \widehat{\Pi}_J^{\perp} L_{K^{1/2}}x, \theta_0\rangle = b_n(L_{K^{1/2}}x) + v_n(L_{K^{1/2}}x)$ for every n and $x \in \mathrm{supp}(P_X^n)$, where*

$$\limsup_{n\to\infty} \sup_{P_Z^n} \sup_{z\in\mathrm{supp}(P_Z^n)} \sup_{\beta_0\in\Theta_1(\mathcal{H})} \frac{|b_n(z)|}{\sqrt{\mu_n}} < +\infty \quad a.s.,$$

and

$$\lim_{n\to\infty} \sup_{P_Z^n} \sup_{z\in\mathrm{supp}(P_Z^n)} \sup_{\beta_0\in\Theta_1(\mathcal{H})} \sup_{t\in\mathbb{R}} \left|\Pr\left[\frac{v_n(z)}{\widehat{\sigma}_n^{\mathsf{adptr}}(z)} \leq t\right] - \Phi(t)\right| = 0,$$

where $\theta_0 = L_{K^{-1/2}}\beta_0$, $\widehat{\Pi}_J^{\perp} = I - \sum_{j=1}^{J(n)} \widehat{\psi}_j \otimes \widehat{\psi}_j$, and $\Phi(\cdot)$ is the CDF of the standard Normal distribution. The probability is over the randomness of both $\{X_i\}_{i=1}^n$ and $\{\varepsilon_i\}_{i=1}^n$.

Furthermore, if $\frac{2r}{(2r-1)(2r+1)} < q < \frac{1}{2}$ *and Assumption (C3) holds, then for every* $\epsilon > 0$, *there exist measurable sets* $\mathcal{X}_1 \subseteq \operatorname{supp}(P_X^1), \mathcal{X}_2 \subseteq \operatorname{supp}(P_X^2), \cdots$ *satisfying* $P_X^1(\mathcal{X}_1), P_X^2(\mathcal{X}_2), \cdots \geq 1-\epsilon$, *such that for any sequence* $x_1 \in \mathcal{X}_1, x_2 \in \mathcal{X}_2, \cdots$, *it holds that*

$$\lim_{n\to\infty} \sup_{\beta_0 \in \Theta_1(\mathcal{H})} \sup_{t \in \mathbb{R}} \left| \Pr\left[\frac{\widehat{\eta}_n^{\mathsf{adptr}}(x) - \eta_0(x)}{\widehat{\sigma}_n^{\mathsf{adptr}}(L_{K^{1/2}} x)} \leq t \right] - \Phi(t) \right| = 0.$$

COROLLARY 7.1. *Impose the same set of assumptions and conditions in Theorem 7.1. Then*

$$\liminf_{n\to\infty} \inf_{P_X^n} \inf_{x \in \operatorname{supp}(P_X^n)} \inf_{\beta_0 \in \Theta_2(\mathcal{H})} \Pr[\eta_0(x) \in \varphi^{\mathsf{adptr}}(x)] \ \geq \ 1-\alpha.$$

Furthermore, if $\frac{2r}{(2r-1)(2r+1)} < q < \frac{1}{2}$ *and Assumption (C3) holds, then the following holds for the choice of* $\omega_n \equiv 0$*: for any* $\epsilon > 0$, *there exist measurable sets* $\mathcal{X}_1 \subseteq \operatorname{supp}(P_X^1), \mathcal{X}_2 \subseteq \operatorname{supp}(P_X^2), \cdots$ *satisfying* $P_X^1(\mathcal{X}_1), P_X^2(\mathcal{X}_2), \cdots \geq 1-\epsilon$, *such that for any sequence* $x_1 \in \mathcal{X}_1, x_2 \in \mathcal{X}_2, \cdots$,

$$\lim_{n\to\infty} \inf_{\beta_0 \in \Theta_1(\mathcal{H})} \Pr[\eta_0(x) \in \varphi^{\mathsf{adptr}}(x)] \ = \ 1-\alpha.$$

7.2. *Rates of convergence.*

THEOREM 7.2. *Impose all assumptions and conditions in Lemma 6.1. Suppose* Λ_n *is constructed as in Eq. (17), where* μ_n *satisfies Eq. (16). Then as* $n \to \infty$,

$$\sup_{P_X^n} \sup_{\beta_0 \in \Theta_1(\mathcal{H})} \mathbb{E}\left[\int \left| \varphi^{\mathsf{adptr}}(x) \right|^2 \mathrm{d}P_X^n(x) \right] \lesssim n^{-\frac{2r}{2r+1}+\delta} + n^{2q-\frac{3}{2}+\delta} + n^{-q(2r-1)+\delta} + n^{1-c}.$$

for any $\delta > 0$, *where* $c \geq 2$ *is the parameter in the failure probability in Lemma 6.1.*

REMARK 7.1. For both Theorems 7.1 and 7.2 to hold with an $n^{-2r/(2r+1)+\delta}$ mean-squared confidence interval width, q must satisfy

$$\frac{2r}{(2r-1)(2r+1)} \ < \ q \ < \ \frac{1}{2}.$$

Provided that $r > (1+\sqrt{2})/2 = 1.207$, the valid range of q is non-empty.

The "effective dimension" d at which $s_d \asymp \mu_n$, is $d \asymp n^{1/(2r+1)}$ when $\mu_n \approx n^{-2r/(2r+1)}$. This means that the "truncation level" J should be significantly larger than d, because $q > \frac{2r}{(2r-1)(2r+1)} > \frac{1}{2r+1}$. This means that, in the anisotropic setting, *both* truncation and Ridge smoothing are necessary to achieve optimal honest and adaptive confidence intervals.

7.3. *Minimax optimality of* φ^{adptr}.

8. Simulation study. [not finished] In this section we empirically validate the four pointwise-CI procedures developed in Sections 4–7, namely the undersmoothed isotropic interval φ^{iso}, the undersmoothed anisotropic interval φ^{ani}, the truncated anisotropic interval φ^{anitr}, and the adaptive-weight interval φ^{adptr}, and we benchmark them against a standard functional principal-component (FPCA) estimator. The controlled simulation of Sections 8.1 and 8.2 is designed to probe three questions: *(i)* does each construction attain its nominal coverage in finite samples, as promised by Corollaries 4.1, 5.1, and 7.1; *(ii)* how do the mean CI widths compare across methods, so that the gains from anisotropy, truncation, and the adaptive weights of Section 7 can be isolated; and *(iii)* how sensitive are these procedures along the axes that our theory flags as delicate, namely the covariance smoothness r_2 in Assumption (C2), the slope regime governing the absolute-summability condition (D1), and the truncation level q in $J = \lfloor n^q \rfloor$ on which Theorem 7.2 imposes a lower bound. Finally, Section 8.3 complements the controlled study with an illustration on real data, the Low Carbon London household electricity-consumption trial, asking whether the relative method-level patterns established in the simulation persist when the population covariance is replaced by the empirical covariance of an actual functional dataset.

8.1. *Experimental setup.* *Data generating process.* Each Monte-Carlo replicate draws n i.i.d. pairs $\{(X_i, Y_i)\}_{i=1}^n$. The input functions are expanded in the cosine basis $\phi_k(t) = \sqrt{2}\cos(k\pi t)$ on $I = [0,1]$, truncated at $K = 200$:

$$X_i(t) = \sum_{k=1}^{K} \xi_{ik}\phi_k(t), \qquad \xi_{ik} \sim \mathcal{N}(0, \theta_k) \text{ independently.}$$

Responses follow $Y_i = \int_0^1 X_i(t)\beta_0(t)\mathrm{d}t + \varepsilon_i$ with $\varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$ and $\sigma = 0.5$. The RKHS $\mathcal{H}$ is taken to be the second-order Sobolev space, whose reproducing kernel has cosine eigenvalues $\mu_k = 2/(k\pi)^4 \asymp k^{-4}$. When the eigenbasis of Σ coincides with the cosine basis (the Aligned design introduced below), the operator $T = L_{K^{1/2}}\Sigma L_{K^{1/2}}$ has eigenvalues $k^{-(4+2r_2)}$ in that basis, giving a spectral-decay exponent $2r = 4 + 2r_2$ that drives the rate constants $2r/(2r+1)$ and $(2r-1)/2r$ in Theorems 7.1–7.2. The Haar and Shifted designs introduce misalignment between the eigenbasis of Σ and that of $L_{K^{1/2}}$, so the eigenvalues of T obey a different design-dependent (but still polynomial) decay law that is not expressible as a closed-form function of r_2 alone.

Covariance designs. We consider three regimes for the population covariance operator Σ of X, each engineered to target a different aspect of Assumptions (B1)–(B2):

- **Aligned**: eigenfunctions coincide with $\{\phi_k\}$ and $\theta_k = k^{-2r_2}$; the population eigenbasis is aligned with the RKHS eigenbasis and (B1)–(B2) hold with $q = 0$.
- **Haar**: eigenfunctions are the Haar wavelets while still $\theta_k = k^{-2r_2}$; the population eigenbasis is misaligned with the RKHS eigenbasis, and (B2) holds only with a nontrivial $q > 0$.
- **Shifted** ($k_0 = 25$): eigenfunctions are $\{\phi_k\}$ but the eigenvalue ordering is scrambled via $\theta_k = (|k - k_0| + 1)^{-2}$, so the leading spectral directions of Σ sit at high frequency, an adversarial case for any procedure that ranks eigendirections by the magnitudes of $\widehat{s}_j$.

For the Aligned and Haar designs we fix the covariance smoothness at $r_2 = 1$; in the Aligned design this places T's decay exponent at $r = 2 + r_2 = 3$, comfortably inside the regime $r > (1+\sqrt{2})/2 \approx 1.207$ in which the valid range of q in Theorem 7.2 is non-empty.

Slope functions. The slope function is represented in the same cosine basis as X,

$$\beta_0(t) = \sum_{k=1}^{K} b_k\,\phi_k(t), \qquad \phi_k(t) = \sqrt{2}\cos(k\pi t), \;\; t \in [0,1],$$

truncated at the same $K = 200$. Because ϕ_k are orthonormal in $L^2([0,1])$, the coefficients coincide with the Fourier-cosine coefficients of β_0, i.e. $b_k = \langle \beta_0, \phi_k\rangle = \int_0^1 \beta_0(t)\phi_k(t)\mathrm{d}t$, and the responses can equivalently be written as $Y_i = \sum_{k=1}^K \xi_{ik}b_k + \varepsilon_i$. To isolate the role of Assumption (D1), we consider three regimes for $\{b_k\}$:

- **Polynomial,** $b_k \propto k^{-2}$: $b_k = 4(-1)^{k-1}/k^2$. Since the transformed coefficient $\langle\theta_0, \phi_k\rangle \asymp b_k/\sqrt{\mu_k} \asymp k^0$, the series $\sum_k |\langle\theta_0, \phi_k\rangle|$ diverges and (D1) *fails*.
- **Polynomial,** $b_k \propto k^{-4}$: $b_k = 4(-1)^{k-1}/k^4$. Now $\langle\theta_0, \phi_k\rangle \asymp k^{-2}$, so (D1) holds.
- **Sparse**: $b_1 = 4$, $b_2 = -2$, and $b_k = 0$ for $k \geq 3$; θ_0 has finite support and (D1) is trivial.

Test functions, replications, and significance levels. Test inputs $\{x_\ell\}_{\ell=1}^{500}$ are drawn once per (covariance design, sample size) configuration from the same distribution as the training inputs, with a fixed random seed independent of the training data, and re-used across all 500 Monte-Carlo replicates of that configuration. Each configuration is averaged over 500 Monte-Carlo replications, with the nominal significance level α swept over 30 values; we display the three levels $1-\alpha \in \{75\%, 85\%, 95\%\}$. Sample sizes are $n \in \{1000, 2000, 4000\}$.

Methods compared. We evaluate five pointwise CI constructions; the truncated and adaptive-weight estimators share a common truncation level $J = \lfloor n^{0.4} \rfloor$ so that their performance is directly comparable.

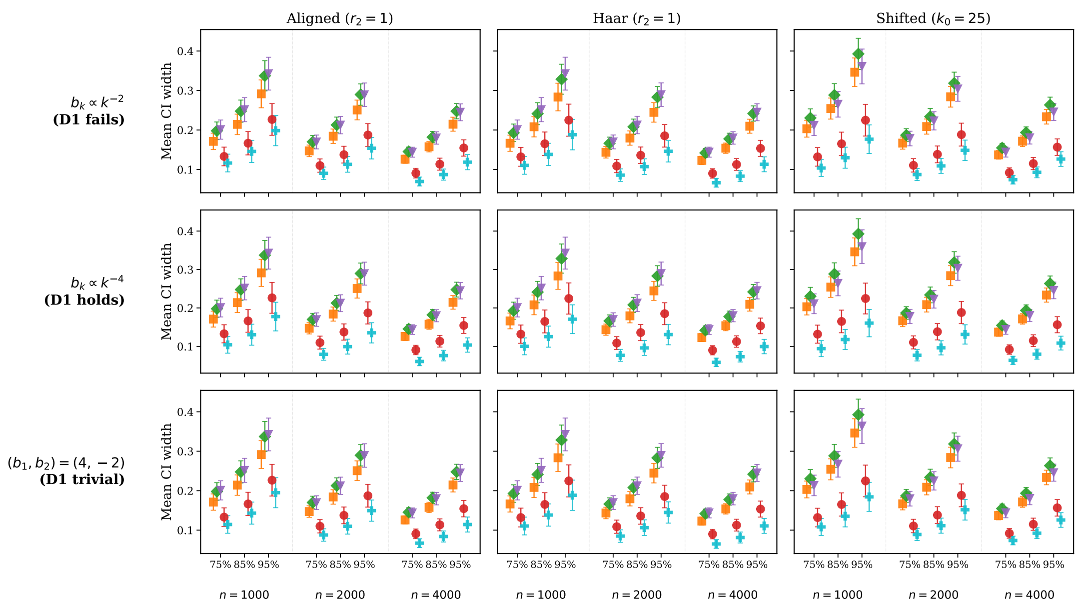
(i) **Aniso**: the undersmoothed anisotropic interval φ^{ani} of Section 5 with $\lambda_{nj} = \sqrt{\widehat{s}_j \lambda_n}$;
(ii) **Iso**: the undersmoothed isotropic interval φ^{iso} of Section 4;
(iii) **FPCA**: a standard baseline that truncates X to its $J = \lfloor\sqrt{n}\rfloor$ leading empirical principal components and uses a Student-t quantile with $n - J$ degrees of freedom;
(iv) **Trunc. Aniso** at $J = \lfloor n^{0.4} \rfloor$: the truncated-anisotropic interval φ^{anitr} of Section 6;
(v) **Adapt. Trunc** at $J = \lfloor n^{0.4} \rfloor$: the adaptive-weight estimator φ^{adptr} of Section 7 with μ_n chosen by 10-fold cross-validation and subsequently scaled by $(\log n)^{-2}$.

Choosing λ_n. The three KRR-type intervals (Aniso, Iso, and Trunc. Aniso) all require an *undersmoothed* scalar penalty λ_n, in the sense that λ_n must decay faster than the rate that is optimal for mean squared estimation error: at the MSE-optimal λ_n the bias of $\langle z, \widehat{\beta}_n\rangle$ does not vanish faster than its standard deviation, and the asymptotic normality used to form the interval breaks down. Standard k-fold cross-validation targets MSE optimality and therefore does not deliver a λ_n of the right order for these intervals. We instead select λ_n on a log-spaced grid of length $L = 2n$ by a Lepski-type adaptive rule with threshold $\kappa = C\sqrt{\log n + \log L}$ and $C = 0.005$, in the spirit of the honest adaptive confidence procedure of Chernozhukov, Chetverikov and Kato (2014); the resulting $\widehat{\lambda}_n$ is then further shrunk by a factor of $(\log n)^{-2}$ to enforce strict undersmoothing.

For Adapt. Trunc the situation is different. The adaptive-weight construction of Section 7 produces an interval whose optimal regularization order in μ_n *coincides* with the MSE-optimal order for the underlying estimation task, so a cross-validation criterion calibrated to estimation error is well-suited to selecting μ_n here. We accordingly pick μ_n by 10-fold cross-validation over a log-spaced grid, and apply the same $(\log n)^{-2}$ shrinkage as a uniform safety margin matching the KRR procedures.

8.2. *Results.* Figure 1 reports two-sided coverage (panel a) and mean CI width (panel b) of the five methods at fixed covariance smoothness $r_2 = 1$. Each panel is a 3×3 grid. Rows vary the slope across the three regimes singled out by Assumption (D1), namely $b_k \propto k^{-2}$ for which (D1) *fails*, $b_k \propto k^{-4}$ for which (D1) is satisfied, and the sparse slope $(b_1, b_2) = (4, -2)$ for which (D1) holds trivially. Columns vary the covariance design (Aligned, Haar, Shifted). Within every cell of either grid, the nine x-positions index the three sample sizes $n \in \{1000, 2000, 4000\}$ paired with the three target levels $\{75\%, 85\%, 95\%\}$, and the six methods are jittered at each tick; error bars are one standard deviation across the 500 Monte-Carlo replicates.

(a) Two-sided coverage; dashed lines mark nominal levels.



(b) Mean CI width on a linear vertical scale.

Fig 1: Two-sided coverage (panel a) and mean CI width (panel b) of the five pointwise CI methods at $r_2 = 1$, with test inputs drawn from the same distribution as the training inputs. The shared legend appears in the top strip. Rows: slope $b_k \propto k^{-2}$ (D1 fails) / $b_k \propto k^{-4}$ (D1 holds) / sparse $(b_1, b_2) = (4, -2)$ (D1 trivial). Columns: covariance design Aligned / Haar / Shifted. Error bars are ± 1 s.d. across 500 Monte-Carlo replicates.

The three intervals introduced in Sections 5–7, namely φ^{ani}, φ^{anitr}, and φ^{adptr}, attain nominal coverage in every cell of Figure 1, with the average absolute distance between observed and nominal two-sided coverage staying below 1% throughout. The slope axis exerts essentially no influence on this calibration: even on the $b_k \propto k^{-2}$ row, where Assumption (D1) provably fails, the coverage error of φ^{anitr} is indistinguishable from that on the (D1)-satisfying rows. The empirical honesty of the proposed methods therefore extends beyond the regime in which Theorems 5.2–7.2 formally apply.

The FPCA baseline tells a different story. On the Aligned and Haar designs it tracks nominal coverage to within 0.2%, but on the Shifted design its coverage drops by 30–40 percentage points at $n = 1000$ and $n = 2000$ across all three slopes, before recovering at $n = 4000$. This is exactly the failure mode the RKHS framework was designed to address: when the population eigenvalues peak away from $k = 1$, ranking empirical principal components by magnitude inverts the ordering of the leading directions at moderate n, and FPCA truncates the wrong subspace. None of the four KRR-type intervals exhibit a comparable failure; their Shifted coverage stays within 1% of nominal at every n.

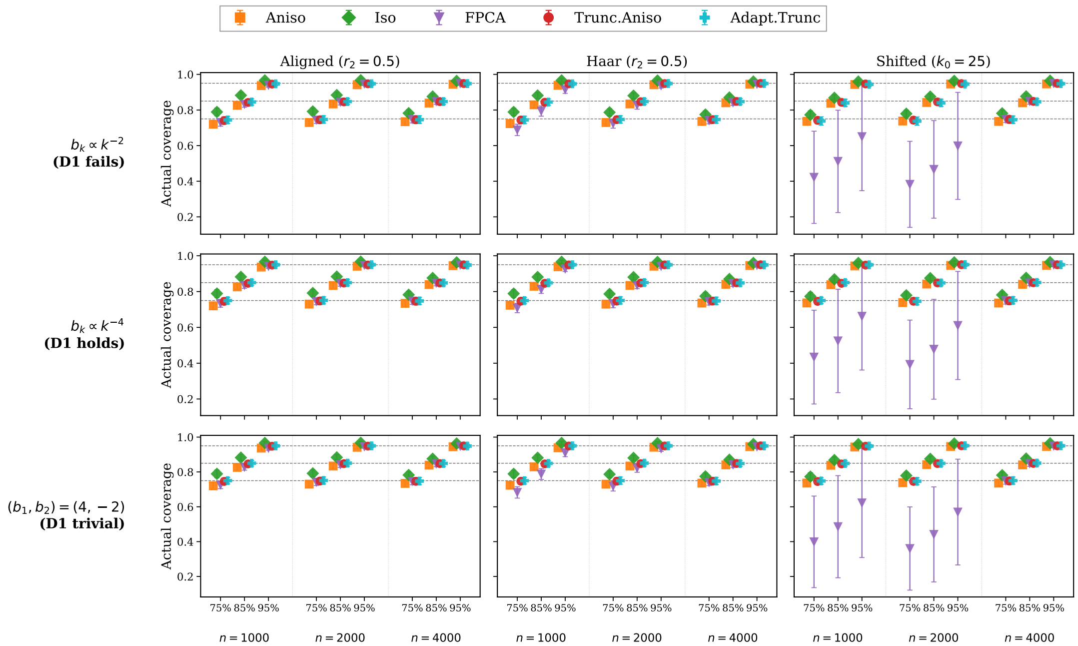
Even outside the Shifted regime, the undersmoothed isotropic interval φ^{iso} is visibly miscalibrated, over-covering by $+1.6\%$–$+3.7\%$ at every $(n, \text{design}, \text{slope})$ combination, with a gap that does not shrink with n. This conservativeness is paid for in width: φ^{iso} is consistently 15–25% wider than φ^{ani} across the grid, an empirical illustration of the cost of using a penalty that is blind to the eigenstructure of T.

On the width dimension, all nine cells obey the same ordering, $\varphi^{\mathsf{adptr}} < \varphi^{\mathsf{anitr}} < \varphi^{\mathsf{ani}} < \varphi^{\mathsf{iso}} \approx$ FPCA, and the adaptive interval φ^{adptr} is the narrowest in every panel while matching or improving on the other methods' coverage. Quantitatively, φ^{adptr} is approximately 25% narrower than φ^{anitr} at the common truncation level $J = \lfloor n^{0.4} \rfloor$, and 45–55% narrower than the un-truncated KRR baselines, an empirical realisation of the variance gain promised by the adaptive weights of Section 7.

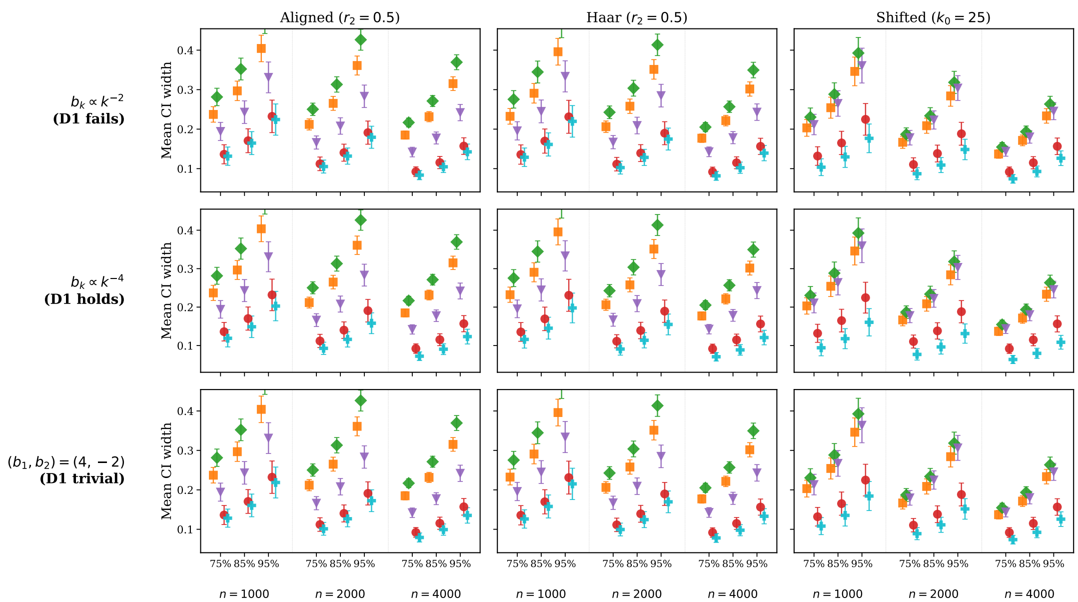
Figures 2 and 3 repeat the experiment at the rougher covariance regime $r_2 = 0.5$ and the smoother regime $r_2 = 2$. The four KRR-type intervals tighten monotonically as r_2 increases, reflecting the easier estimation problem under smoother covariance: averaged over the Aligned and Haar designs and the three slopes, the mean width of φ^{ani} drops from 0.27 at $r_2 = 0.5$ to 0.19 at $r_2 = 1$ and 0.12 at $r_2 = 2$, with φ^{iso} following the same trajectory. The truncation-based intervals φ^{anitr} and φ^{adptr} shrink more modestly, as their truncation cap already controls variance. FPCA, by contrast, is essentially insensitive to r_2 in width (its $J = \lfloor \sqrt{n} \rfloor$ truncation discards the additional smoothness), and its coverage on the Aligned and Haar designs is mildly under-calibrated at $r_2 = 0.5$ (mean error $\approx 1.7\%$) while remaining nearly perfect at $r_2 \in \{1, 2\}$. The catastrophic failure of FPCA on the Shifted design persists at both r_2 values, since the Shifted covariance is itself r_2-independent. The qualitative method ordering of Figure 1 is otherwise unchanged.

To isolate the role of the truncation level, Figure 4 restricts attention to the $b_k \propto k^{-4}$ slope and compares the two truncation-based intervals, φ^{anitr} and φ^{adptr}, at three common truncation levels $J \in \{\lfloor n^{0.2} \rfloor, \lfloor n^{0.3} \rfloor, \lfloor n^{0.4} \rfloor\}$. Columns correspond to $r_2 \in \{0.5, 1, 2\}$ and rows to the Aligned and Haar covariance designs (the Shifted design is omitted here as its eigenstructure is largely orthogonal to the truncation choice). Because the two rows share a common y-axis on each panel of width and coverage, the dependence of method behaviour on r_2 can be read off horizontally.

Figure 4 contains a direct empirical check of the theoretical lower bound on the truncation exponent in Theorem 7.2, $q > \frac{2r}{(2r-1)(2r+1)}$. In the Aligned designs the operator T has decay exponent $r = 2 + r_2$, so the theoretical lower bound evaluates to $q_{\min} \approx 0.208$, 0.171, and 0.127 at $r_2 = 0.5$, 1, and 2 respectively. Only at $r_2 = 0.5$ does $q = 0.2$ fall below this threshold, and the figure shows exactly this point of breakdown: averaged over n, target level, and



(a) Two-sided coverage; dashed lines mark nominal levels.

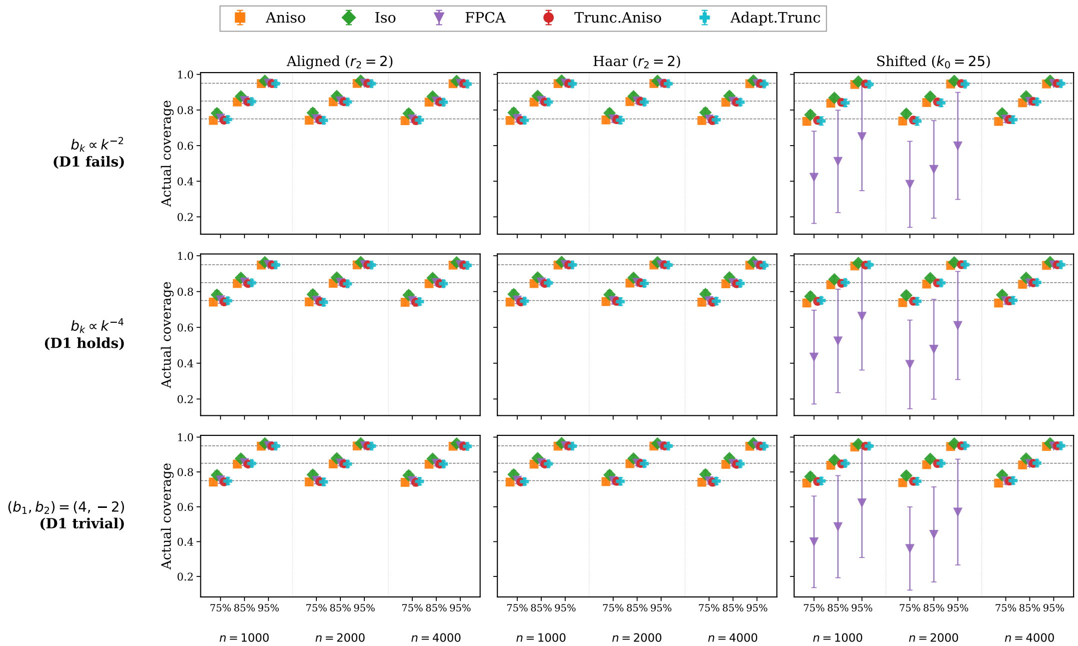


(b) Mean CI width on a linear vertical scale.

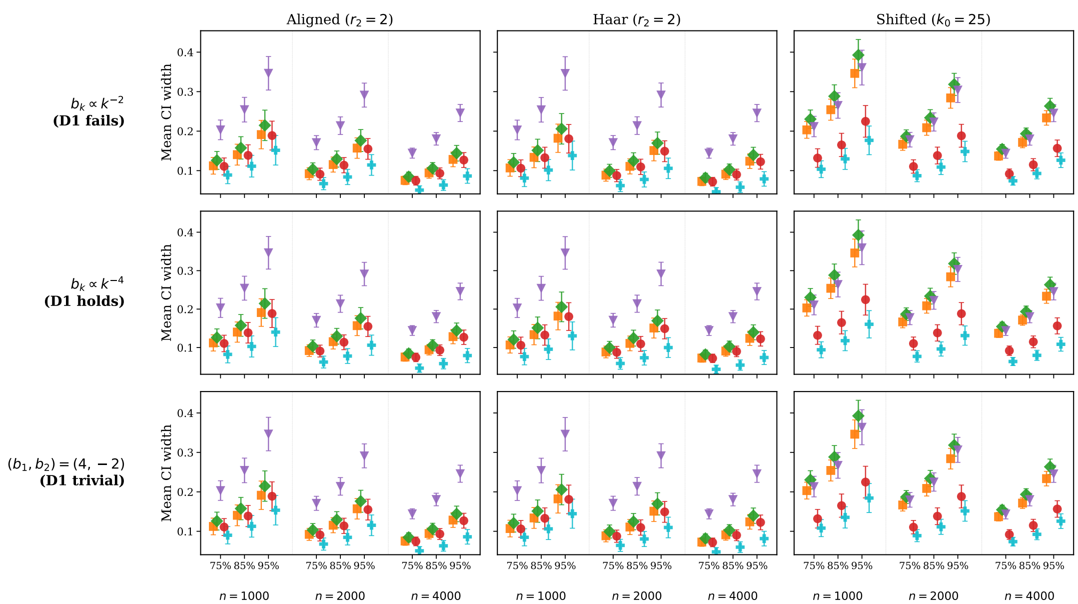
Fig 2: Same layout as Figure 1 at the rougher covariance regime $r_2 = 0.5$.

the two designs, the coverage error of both φ^{anitr} and φ^{adptr} at $q = 0.2$ jumps to $\approx 1.1\%$ in the $r_2 = 0.5$ column, an order of magnitude above the $\leq 0.2\%$ errors at $q \in \{0.3, 0.4\}$ in the same column, while at $r_2 \in \{1, 2\}$ all three q values produce uniformly small coverage errors below 0.5%.

The width behaviour separates the two methods most clearly at moderate truncation levels. At the smallest $q = 0.2$ both methods are insensitive to r_2, with widths changing by less than



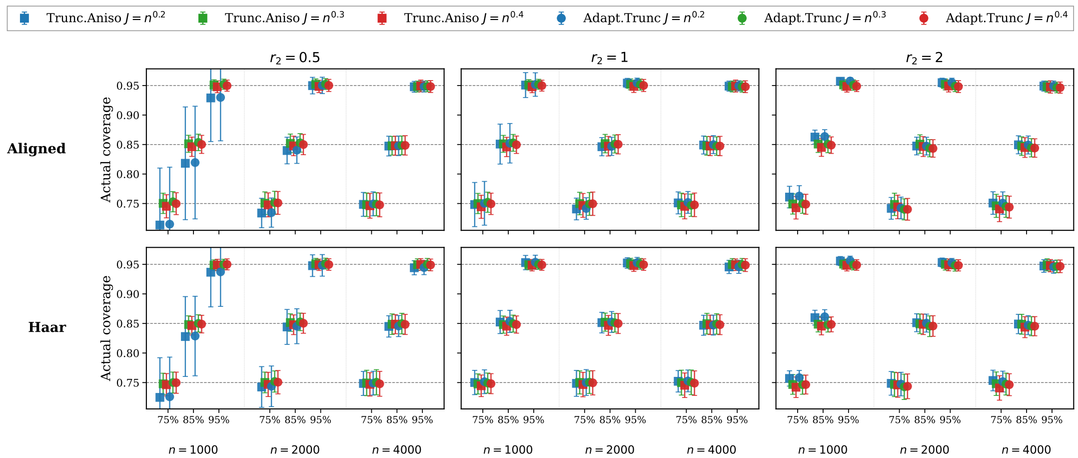
(a) Two-sided coverage; dashed lines mark nominal levels.



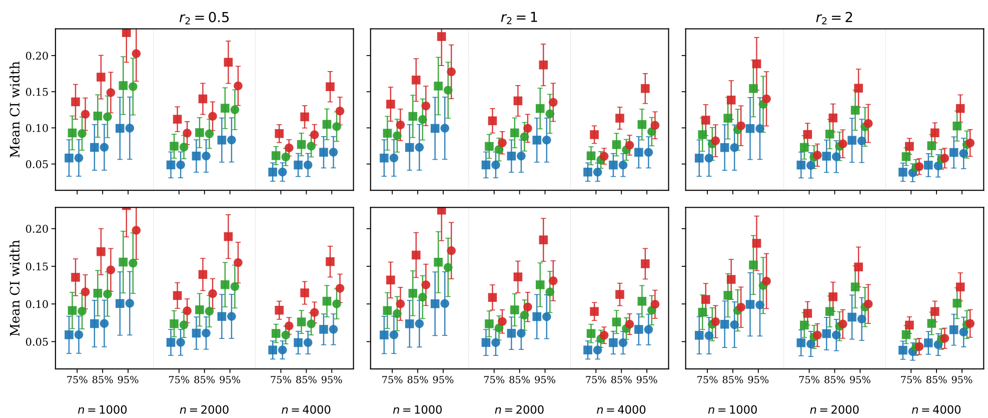
(b) Mean CI width on a linear vertical scale.

Fig 3: Same layout as Figure 1 at the smoother covariance regime $r_2 = 2$.

3% across $r_2 \in \{0.5, 1, 2\}$, and the two methods give essentially identical widths, since the adaptive correction provides no benefit once the truncation level is small enough to dominate the variance-bias trade-off. At $q = 0.3$, Trunc. Aniso width remains nearly flat in r_2 (a 2% decrease) whereas Adapt. Trunc width drops by roughly 21%. At $q = 0.4$ both widths fall with r_2, but Adapt. Trunc's decrease (34%) is substantially larger than Trunc. Aniso's (20%). Consequently, the ratio of Adapt. Trunc to Trunc. Aniso width falls steadily with r_2: at $q =$



(a) Two-sided coverage; dashed lines mark nominal levels.



(b) Mean CI width on a linear vertical scale.

Fig 4: Truncation-level sensitivity at slope $b_k \propto k^{-4}$ with input-distribution test points: φ^{anitr} (squares) and φ^{adptr} (circles) at $J = \lfloor n^q \rfloor$ for $q \in \{0.2, 0.3, 0.4\}$ (blue / green / red). Columns: $r_2 \in \{0.5, 1, 2\}$. Rows: covariance design Aligned / Haar. Within each cell, the nine x-positions index $n \in \{1000, 2000, 4000\}$ paired with target levels $\{75\%, 85\%, 95\%\}$; error bars are ± 1 s.d. across 500 Monte-Carlo replicates.

0.4 it goes from 0.83 at $r_2 = 0.5$ to 0.72 at $r_2 = 1$ and 0.68 at $r_2 = 2$. The advantage of φ^{adptr} over φ^{anitr} is therefore concentrated at the larger q values, where there is more variance for the adaptive weights to tame, and grows with r_2, where the smoother covariance gives them more leverage.

8.3. *Real-data illustration.* The aim of this subsection is to check whether the method-level patterns established in our controlled simulation, in particular the honesty of the proposed intervals and the width ordering across the five methods, persist when the population covariance is replaced by the empirical covariance of a real functional dataset. Since the true slope β_0 is unobserved on real data, the target $\eta(x_{\text{new}}) = \langle x_{\text{new}}, \beta_0 \rangle$ has no ground-truth value at any actual covariate, so we cannot measure coverage directly. We therefore frame what follows as an *illustration* of method-level CI centres and widths under a real-data covariance, rather than as a coverage validation; coverage validation against known ground truth is the role of Section 8.2.

The Low Carbon London (LCL) trial is a residential smart-meter study run by UK Power Networks between November 2011 and February 2014. Approximately 5,500 London households participated, with their domestic electricity consumption recorded at 30-minute resolution over the trial window. The participating households were split into two cohorts: a *Std* (control) cohort that kept their standard flat-rate electricity tariff throughout the trial, and a *ToU* (treatment) cohort that was switched onto a dynamic time-of-use tariff starting on 2013-01-01. The resulting dataset is publicly available from UK Power Networks and has become a standard benchmark for residential electricity-consumption modelling. We restrict attention to the Std cohort and treat each household's *pre-trial weekly load profile* (the household's typical half-hourly consumption pattern over a calendar week, observed before the dynamic-tariff intervention) as a functional covariate, and its *post-trial average daily consumption* as the scalar response. This setup mirrors a typical forecasting task: predict the household's average level of electricity demand from its historical consumption pattern.

We apply the proposed pointwise CI methods to the Std cohort just described, splitting each household's half-hourly smart-meter readings into a *pre-trial* segment (timestamps before 2013-01-01) and a *post-trial* segment (timestamps on or after 2013-01-01).

The covariate is constructed from the pre-trial segment as follows. For each household i and each (d,h) pair with $d \in \{\text{Mon}, \text{Tue}, \ldots, \text{Sun}\}$ and $h \in \{1, \ldots, 48\}$, let $N_i(d,h)$ be the number of pre-trial half-hourly readings of household i falling in day-of-week d and half-hour-of-day h, and let $S_i(d,h)$ be their sum. The raw weekly profile is

$$x_i^{\text{raw}}(d,h) \;=\; \begin{cases} S_i(d,h)/N_i(d,h) & \text{if } N_i(d,h) > 0, \\ \text{row mean} & \text{if } N_i(d,h) = 0 \text{ (mean across other bins of household } i), \end{cases}$$

giving a $p = 7 \times 48 = 336$-dimensional vector indexed by (d,h). Households with no pre-trial readings at all are discarded. We then trim outliers in the profile-norm distribution: writing $\overline{x}^{\text{raw}} = n^{-1} \sum_i x_i^{\text{raw}}$ for the sample mean, we retain only those households whose centred-norm $\|x_i^{\text{raw}} - \overline{x}^{\text{raw}}\|_2$ lies in the $[20\%, 80\%]$ quantile range across the sample. The scalar response is the post-trial average daily total consumption,

$$T_i \;=\; \frac{1}{\#\text{days}_i^{\text{post}}} \sum_{t \in \text{post-trial}} \text{kWh}_{i,t},$$

and households with $T_i \le 0$ are dropped. After all filtering steps we are left with $n = 2640$ households. The final training covariate X_i and response Y_i are obtained by column-centring the trimmed predictor and centring the log response:

$$X_i(d,h) \;=\; x_i^{\text{raw}}(d,h) - \frac{1}{n} \sum_{i'=1}^{n} x_{i'}^{\text{raw}}(d,h), \qquad Y_i \;=\; \log T_i - \frac{1}{n} \sum_{i'=1}^{n} \log T_{i'}.$$

The RKHS structure is fixed by a preliminary model-selection step: we evaluate the anisotropic Lepski estimator by 5-fold cross-validation on a grid of (basis, K, r) triples and adopt the winning configuration, the Haar wavelet basis with $K = 32$ and $r = 1.5$, for the cross-method comparison below. Lepski's threshold uses $C = 0.005$, matching the simulation.

In place of held-out real households (for which $\eta(x_{\text{new}})$ would have no observed value), we evaluate the methods at a fixed pool of $N_{\text{test}} = 100$ *synthetic test covariates* whose spectral structure matches the LCL training distribution. Letting $\widehat{\Sigma} = n^{-1} X^\top X$ be the empirical covariance operator of the centred training features and $\widehat{\Sigma} = U \operatorname{diag}(w_1, \ldots, w_p) U^\top$ its spectral decomposition with w_j floored at 0 (to guard against numerical negatives), we draw

$$x_\ell^{\text{test}} \;=\; U \cdot \operatorname{diag}\big(\sqrt{w_1}, \ldots, \sqrt{w_p}\big) \cdot Z_\ell, \qquad Z_\ell \overset{\text{iid}}{\sim} \mathcal{N}(0, I_p), \quad \ell = 1, \ldots, N_{\text{test}},$$

so that $x_\ell^{\text{test}} \sim \mathcal{N}(0, \widehat{\Sigma})$. The random seed is fixed so that the same 100 test covariates are used by every method. We emphasise that the test set consists of *covariates only*: no synthetic response y_ℓ^{test} is generated and no ground-truth $\eta(x_\ell^{\text{test}})$ is assumed. For each test covariate x_ℓ^{test} and each method, we fit $\widehat{\beta}$ on the $n = 2640$ LCL training pairs $\{(X_i, Y_i)\}_{i=1}^n$ and report the resulting pointwise CI for $\eta(x_\ell^{\text{test}})$, centred at the method's own point estimate $\widehat{\eta}(x_\ell^{\text{test}}) = \langle x_\ell^{\text{test}}, \widehat{\beta} \rangle$. Figure 5 compares those CIs across methods at the same 100 test covariates.

The five methods reported in the figure are φ^{ani}, φ^{iso}, FPCA at $J = \lfloor \sqrt{n} \rfloor$, and φ^{anitr} and φ^{adptr} at the common default truncation level $J = \lfloor n^{0.3} \rfloor$. We take $q = 0.3$ as the default truncation level because Figure 4 shows it is the smallest truncation level whose coverage stays honest across all simulated covariance-smoothness levels $r_2 \in \{0.5, 1, 2\}$.

Panel (a) of Figure 5 draws side-by-side 95% intervals for fifteen representative test functions, quantile-spaced by the Aniso point estimate, so that the horizontal axis covers a range of $\widehat{\eta}(x_\ell^{\text{test}})$ values from approximately -0.9 to $+0.9$ in log-kWh units. Across the middle of this range the five method intervals overlap pairwise, indicating broad agreement on the location of $\eta(x_\ell^{\text{test}})$ up to their respective widths. Toward the tails, in particular at the largest $\widehat{\eta}$ test curve and to a lesser extent at the smallest, the intervals separate a bit.

Panel (b) summarises the half-width distribution across all 100 test inputs as a grouped boxplot, with three boxes per method corresponding to the three nominal levels $\{75\%, 85\%, 95\%\}$. At every nominal level and across the bulk of the test-curve distribution, Adapt. Trunc produces the narrowest intervals and FPCA the widest, with Trunc. Aniso and Iso lying in between, matching the width ordering observed in the controlled simulation (cf. Figure 1). The consistency of this ordering between the simulation and the LCL-trained model suggests that the relative efficiency advantages of the truncated and adaptive estimators are not artefacts of a particular simulated covariance but carry over to a real spectral structure, providing the empirical confirmation of the simulation findings that this subsection set out to provide.

We close with a recommendation for practitioners. As a default choice we recommend Adapt. Trunc with $J = \lfloor n^{0.3} \rfloor$: across both the controlled simulation and the LCL data it produces the narrowest intervals while keeping coverage at the nominal level. One caveat is that at test inputs in the extreme tails of the prediction distribution, Adapt. Trunc returns a point estimate with smaller absolute value than the four other methods (panel (a) of Figure 5): at the most positive test curve the other four methods predict $\widehat{\eta}$ in the range $+0.82$ to $+0.89$ while Adapt. Trunc returns $+0.79$, and at the most negative test curve the other four cluster near -0.93 while Adapt. Trunc returns -0.89. When accuracy on tail covariates is the priority, the Aniso or Trunc. Aniso interval is the safer choice.

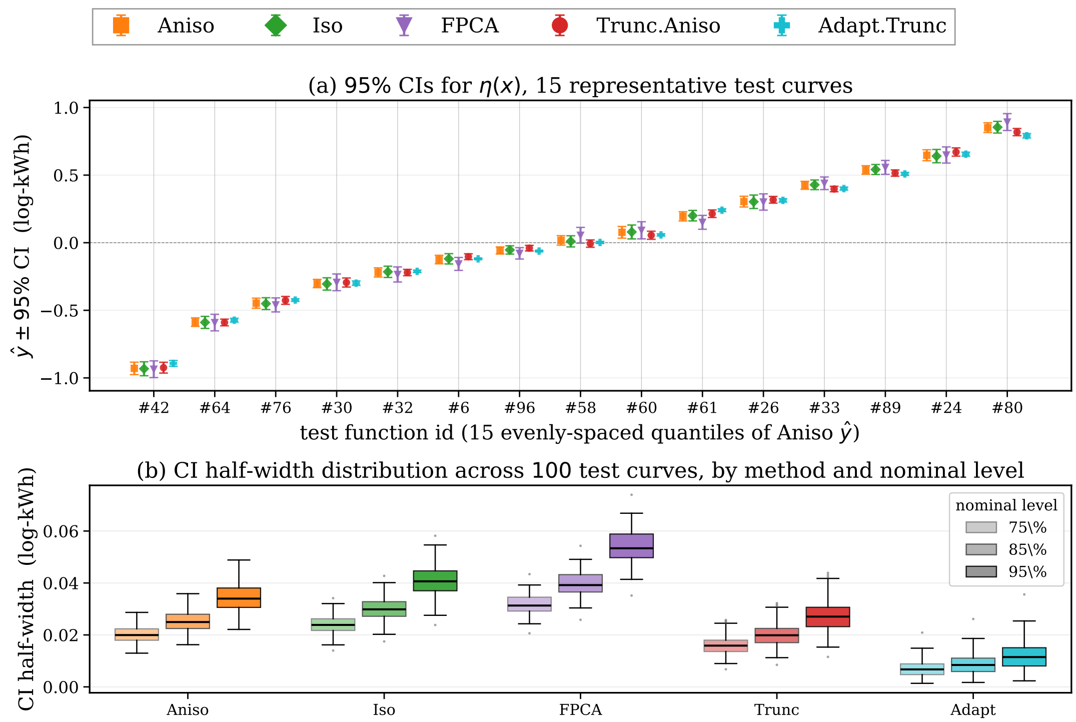


Fig 5: Pointwise CIs for $\eta(x) = \langle x, \beta_0 \rangle$ from five methods trained on $n = 2640$ LCL Std households ($p = 336$ time points, Haar basis with $K = 32$ and $r = 1.5$, $C_{\text{Lepski}} = 0.005$). Trunc.Aniso and Adapt.Trunc share the default truncation level $J = \lfloor n^{0.3} \rfloor$. Test inputs are 100 Gaussian-process draws from $\mathcal{N}(0, \widehat{\Sigma}_{\text{train}})$. The shared method legend appears in the top strip. Panel (a): side-by-side 95% CIs for the fifteen test functions chosen at evenly-spaced quantiles of the Aniso point estimate. Panel (b): CI half-width distribution across all 100 test functions, grouped by method and by nominal level $\{75\%, 85\%, 95\%\}$ (darker shading $=$ higher nominal level); box hinges and whisker conventions are the matplotlib defaults.

APPENDIX A: PROOFS

For brevity we assume $\mathbb{E}[x] = 0$, $\gamma_0 = 0$ and $X_i' = X_i$ throughout all proofs, which is conventional in the analysis of linear functional regression in the literature (Cai and Yuan, 2012; Yuan and Cai, 2010). For all *lower bound* proofs, we also assume $L_{K^{1/2}} = L_{K^{-1/2}}$ is the identity operator to simplify the proofs. For notational brevity, we many times drop the identity operator symbol I, such as writing $T_n + \lambda$ for $T_n + \lambda I$, whenever there is no ambiguity.

A.1. Some technical lemmas for results in Sec. 4. We first establish a few technical lemmas which are important to our proof of later results.

LEMMA A.1. *Suppose Assumptions (A1), (A3) and (B3) hold with $\kappa < \frac{1}{2} - \frac{1}{4r}$, and λ_n satisfies Eq. (9). Then for any $\delta \in (0, 1/2]$ there exists a constant $n_0 \in \mathbb{N}$ depending only on δ and other parameters in assumptions, such that for all $n \geq n_0$, with probability at least $1 - n^{-1}$,*

$$(1-\delta)(T+\lambda_n) \preceq T_n + \lambda_n \preceq (1+\delta)(T+\lambda_n)$$

where $T_n = \frac{1}{n}\sum_{i=1}^n Z_i \otimes Z_i$, $T = \mathbb{E}[Z \otimes Z]$, $Z_i = L_{K^{1/2}} X_i$, $Z = L_{K^{1/2}} Z$, and for two self-adjoint operators A, B, $A \preceq B$ if $\langle x, Ax\rangle \leq \langle x, Bx\rangle$ for any square integrable on function $x : I \to \mathbb{R}$. Furthermore, for any $\delta > 0$, there exists a constant $C < +\infty$ depending only on parameters in the assumptions, such that for all $n \geq 3$, with probability $1 - n^{-1}$ it holds that

$$\left\|(T+\lambda_n)^{-1/2}(T - T_n)(T+\lambda_n)^{-1/2}\right\|_{\mathrm{op}} \leq C \times n^{-\frac{1-2\kappa-1/2r}{2}+\delta}.$$

We remark that similar versions of Lemma A.1 have appeared, in different forms, in classical analysis of kernel ridge regression estimators and RKHS approaches for linear functional regression. See e.g. the upper bound on $\mathcal{S}_2(\lambda, z)$ in (Caponnetto and De Vito, 2007), the second inequality in the proof of Theorem 5 in (Smale and Zhou, 2007), Theorem 2 of (Hsu, Kakade and Zhang, 2014), or Lemma 2 of (Cai and Yuan, 2012). Nevertheless, in our applications the choice of λ_n is significantly smaller than conventional choices (for inference purposes), and when $\lambda_n = o(n^{-1})$ existing arguments cannot be directly applied, without additional conditions such as Assumption (B3) for some $\kappa < 1/2$. Therefore, we give the complete proof of Lemma A.1 below.

PROOF OF LEMMA A.1. We use Hoeffding's inequality for bounded self-adjoint operators, as developed and stated in (Minsker, 2017). For every $i \in [n]$, define

$$W_i := (T+\lambda_n)^{-1/2} Z_i = (T+\lambda_n)^{-1/2} L_{K^{1/2}} X_i.$$

This gives rise to

$$\frac{1}{n}\sum_{i=1}^n W_i \otimes W_i = (T+\lambda_n)^{-1/2} T_n (T+\lambda_n)^{-1/2}; \tag{18}$$

$$\mathbb{E}[W_i \otimes W_i] = (T+\lambda_n)^{-1/2} T (T+\lambda_n)^{-1/2}, \quad \forall i \in [n]. \tag{19}$$

By Assumptions (B1) and (B3), it holds almost surely that

$$\|W_i \otimes W_i\|_{\mathrm{op}} = \left\| \sum_{j,k=1}^{\infty} \frac{\langle Z_i, \psi_j\rangle \langle Z_i, \psi_k\rangle}{\sqrt{s_j+\lambda_n}\sqrt{s_k+\lambda_n}} \psi_j \otimes \psi_k \right\|_{\mathrm{op}}$$

$$\leq \sum_{j=1}^{\infty} \frac{\langle Z_i, \psi_j \rangle^2}{s_j + \lambda_n} \leq \sum_{j=1}^{\infty} \frac{C_\nu s_j n^{2\kappa}}{s_j + \lambda_n} \lesssim n^{2\kappa + 1/2r + \delta}, \tag{20}$$

where the last inequality holds because $\lambda_n \gtrsim n^{-1-\delta}$ for any $\delta > 0$.

Let

$$A_n := \sum_{i=1}^{n} \mathbb{E}[(W_i \otimes W_i)^2] = n \times \sum_{j,k=1}^{\infty} \frac{\mathbb{E}[\langle Z, Z \rangle \langle Z, \psi_j \rangle \langle Z, \psi_k \rangle]}{(s_j + \lambda_n)(s_k + \lambda_n)} \psi_j \otimes \psi_k.$$

It is easy to verify that $\|A_n\|_{\mathrm{op}} \geq n\mathbb{E}[\langle Z, \psi_1 \rangle^4]/(s_1 + \lambda_n) = PD :\geq n s_1^2/(s_1 + \lambda_n) \geq c_s n/2$ for sufficiently large n. On the other hand, Because $\langle Z, Z \rangle \leq 1$ almost surely thanks to Assumption (A3), it holds that

$$\mathrm{tr}(A_n) \leq n \times \sum_{j=1}^{\infty} \frac{s_j}{(s_j + \lambda_n)^2} \lesssim n^{(2r+1)/r}, \tag{21}$$

where the last inequality holds because $s_j \asymp j^{-2r}$ thanks to Assumption (A1), and $\lambda_n \gtrsim n^{-2}$. Invoking (Minsker, 2017, Theorem 3.1), for any $\delta > 0$, it holds with probability $1 - n^{-1}$ that

$$\left\| \frac{1}{n} \sum_{i=1}^{n} W_i \otimes W_i - \mathbb{E}[W_i \otimes W_i] \right\|_{\mathrm{op}} \lesssim n^{-\frac{1-2\kappa-1/2r-\delta}{2}} \log n = o(1).$$

Subsequently, incorporating Eqs. (18,19), the above inequality implies

$$\left\| (T + \lambda_n)^{-1/2} (T - T_n) (T + \lambda_n)^{-1/2} \right\|_{\mathrm{op}} \lesssim n^{-\frac{1-2\kappa-1/2r}{2} + \delta} = o(1).$$

This proves Lemma A.1, by selecting n_0 sufficiently large such that the right-hand side of the above inequality is upper bounded by δ for any $\delta > 0$. $\square$

For the purpose of the next lemma, for any $z = L_{K^{1/2}} x$ for some $x \in \mathcal{X}$ and $Z_i = L_{K^{1/2}} X_i$, define

$$g(z, Z_i) := \langle z, (T_n + \lambda_n)^{-1} Z_i \rangle, \quad \overline{g}(z, Z_i) := \langle z, (T + \lambda_n)^{-1} Z_i \rangle.$$

The following lemma derives probabilistic convergence between g, $\overline{g}$, and their expectations.

LEMMA A.2. *Suppose the same set of conditions in Lemma A.1 hold, with Assumption (B3) being strengthened to $\kappa < \frac{1}{6} - \frac{1}{4r}$. Then it holds that*

$$\sup_{z \in L_{K^{1/2}} \mathcal{X}} \left| \frac{1}{n} \sum_{i=1}^{n} [g(z, Z_i) - \overline{g}(z, Z_i)]^2 \right| \xrightarrow{p} 0;$$

$$\sup_{z \in L_{K^{1/2}} \mathcal{X}} \left| \frac{1}{n} \sum_{i=1}^{n} \overline{g}(z, Z_i)^2 - \mathbb{E}[\overline{g}(z, Z)^2] \right| \xrightarrow{p} 0.$$

PROOF OF LEMMA A.2. Throughout this proof, we write $w := (T + \lambda)^{-1/2} z$ and $W_i := (T_n + \lambda)^{-1/2} Z_i$ for notational simplicity, where $\lambda = \lambda_n$. By Assumptions (B1) and (B3), for sufficiently large n it holds uniformly for almost every $x \in \mathcal{X}$ and $z = L_{K^{1/2}} x$ that

$$\|w\|_2^2 \leq \sum_{j=1}^{\infty} \frac{\langle z, \psi_j \rangle^2}{s_j + \lambda_n} \leq \sum_{j=1}^{\infty} \frac{C_\nu s_j n^{2\kappa}}{s_j + \lambda_n} \lesssim n^{2\kappa + 1/2r + \delta} \tag{22}$$

for any $\delta > 0$.

Define $\Delta_i := g(z, Z_i) - \overline{g}(z, Z_i)$. Then

$$
\begin{aligned}
\Delta_i &= \langle z, ((T_n + \lambda)^{-1} - (T + \lambda)^{-1}) Z_i \rangle \\
&= \langle z, (T + \lambda)^{-1}((T + \lambda)(T_n + \lambda)^{-1} - I) Z_i \rangle \\
&= \langle z, (T + \lambda)^{-1}(T_n - T)(T_n + \lambda)^{-1} Z_i \rangle .
\end{aligned}
$$

Subsequently,

$$
\begin{aligned}
\frac{1}{n}\sum_{i=1}^{n} \Delta_i^2 &= \langle z, (T + \lambda)^{-1}(T_n - T)(T_n + \lambda)^{-1} T_n (T_n + \lambda)^{-1}(T_n - T)(T + \lambda)^{-1} z \rangle \\
&= \langle w, (T + \lambda)^{-1/2}(T_n - T)(T_n + \lambda)^{-1} T_n (T_n + \lambda)^{-1}(T_n - T)(T + \lambda)^{-1/2} w \rangle \\
&\le \langle w, (T + \lambda)^{-1/2}(T_n - T)(T_n + \lambda)^{-1}(T_n - T)(T + \lambda)^{-1/2} w \rangle \\
&\le \|(T + \lambda)^{-1/2}(T_n - T)(T_n + \lambda)^{-1/2}\|_{\mathrm{op}} \|w\|_2^2 \\
&\overset{(a)}{=} O_P\left(n^{-\frac{1-2\kappa-1/2r}{2}+\delta}\right) \times O(n^{2\kappa+1/2r+\delta}) \ \xrightarrow{p}\ 0, \qquad (23)
\end{aligned}
$$

where Eq. (a) holds by invoking Lemma A.1 and incorporating Eq. (22), and noting that

$$
\begin{aligned}
&\|(T + \lambda)^{-1/2}(T_n - T)(T_n + \lambda)^{-1/2}\|_{\mathrm{op}} \\
&\qquad \le \|(T + \lambda)^{-1/2}(T_n - T)(T + \lambda)^{-1/2}\|_{\mathrm{op}} \cdot \|(T + \lambda)^{1/2}(T_n + \lambda)^{-1/2}\|_{\mathrm{op}},
\end{aligned}
$$

where the first term can be upper bounded directly using Lemma A.1, and the second term can be upper bounded by noting that

$$
\|(T + \lambda)^{-1/2}(T_n - T)(T_n + \lambda)^{-1/2}\|_{\mathrm{op}} \le \epsilon \implies (1-\epsilon)(T+\lambda) \preceq T_n + \lambda \preceq (1+\epsilon)(T+\lambda)
$$

and subsequently $\|(T + \lambda)^{1/2}(T_n + \lambda)^{-1/2}\|_{\mathrm{op}} \le 1/\sqrt{1-\epsilon}$. This proves the first property, because Eq. (22) is uniform almost everywhere in z. This proves the first property, because Eq. (22) is uniform in z.

For the second identity, because $\{g(z, Z_i)^2\}_{i=1}^n$ are i.i.d. random variables, the Chebyshev's inequality asserts that for any $t > 0$,

$$
\Pr\left[\left|\frac{1}{n}\sum_{i=1}^{n} \overline{g}(z, Z_i)^2 - \mathbb{E}[g(z, Z)^2]\right| > t\mathbb{E}[g(z, Z)^2]\right] \le \frac{\mathbb{E}[g(z, Z)^4]}{nt^2(\mathbb{E}[g(z, Z)^2])^2} \le \frac{C_p}{nt^2}, \qquad (24)
$$

where the last inequality holds thanks to Assumption (B2). Because the right-hand side of Eq. (24) is uniform in almost every z, This implies that for almost every z,

$$
\frac{1}{n}\sum_{i=1}^{n} \overline{g}(z, Z_i)^2 \ \xrightarrow{p}\ \mathbb{E}[g(z, Z)^2],
$$

This completes the proof.

YW: check if this statement is accurate? □

LEMMA A.3. *Assume the same set of conditions in Lemmas A.1 and A.2. Suppose also that $\kappa < 1/4 - 1/(4r)$ in Assumption (B3). Let $\varepsilon_i = Y_i - \langle X_i, \beta_0 \rangle$. Then for any $x \in \mathcal{X}$ and $z = L_{K^{1/2}} x$, as $n \to \infty$,*

$$
\frac{1}{\widetilde{\sigma}_n(z)\sqrt{n}} \sum_{i=1}^{n} \varepsilon_i g(z, Z_i) \ \xrightarrow{d}\ \mathcal{N}(0, 1),
$$

where $Z_i = L_{K^{1/2}} X_i$, $g(z, Z) = \langle z, (T_n + \lambda_n I)^{-1} Z \rangle$ and $\widetilde{\sigma}_n(z)^2 = n^{-1}\sum_{i=1}^{n} \sigma_\varepsilon^2 g(z, Z_i)^2$.

PROOF OF LEMMA A.3. Define $\overline{g}(z,Z) := \langle z, (T+\lambda_n I)^{-1}Z\rangle$. Let Z, Z' be arbitrary in the support of $P_{\mathcal{Z}}$ induced by $P_{\mathcal{X}}$ with the understanding that $Z = L_{K^{1/2}}X$, and for every $j\geq 1$ define $\alpha_j := \langle Z, \psi_j\rangle$, $\alpha_j' := \langle Z', \psi_j\rangle$, where $\{\psi_j\}_{j=1}^\infty$ is the orthonormal basis of the population covariance operator $T = \mathbb{E}[Z\otimes Z]$. Thanks to Assumption (B3), it holds almost surely that $|\alpha_j|, |\alpha_j'| \leq C_\nu n^\kappa \sqrt{s_j}$ for every $j\geq 1$. Subsequently, with probability 1 over $Z, Z' \sim P_{\mathcal{Z}}$,

$$\left|\langle Z, (T+\lambda_n)^{-1}Z'\rangle\right| \leq C_\nu n^{2\kappa}\sum_{j=1}^\infty \frac{s_j}{s_j+\lambda_n} \lesssim n^{2\kappa}\times \int_1^\infty \frac{j^{-2r}\mathrm{d}j}{j^{-2r}+\lambda_n} \lesssim n^{2\kappa+1/(2r)+\delta}$$

for any $\delta>0$, where the last inequality holds because $\lambda_n n\to 0$ and $\lambda_n n^{1+\delta}\to\infty$ for any $\delta > 0$.

For any z in the support of $P_{\mathcal{Z}}$, let $\varsigma(z)^2 := \mathbb{E}[\langle z, (T+I)^{-1}Z\rangle^2]$. Because $P_{\mathcal{Z}}$ is non-degenerate (see Assumption (B3)), we have that $\varsigma(z)^2 > 0$ for almost every z. Consequently,

$$\begin{aligned}\lim_{n\to\infty}\frac{1}{\sqrt{n}}\frac{\mathbb{E}[|\langle z,(T+\lambda_n)^{-1}Z\rangle|^3]}{\mathbb{E}[\langle z,(T+\lambda_n)^{-1}Z\rangle^2]^{3/2}} &\leq \lim_{n\to\infty}\frac{n^{2\kappa+1/(2r)+\delta}}{\sqrt{n}}\times\frac{1}{\mathbb{E}[\langle z,(T+\lambda_n)^{-1}Z\rangle^2]^{1/2}}\\ &\overset{*}{\leq} \lim_{n\to\infty} n^{2\kappa+(1/2r)+\delta-1/2}\times\sigma^{-1} = 0,\end{aligned}$$

where Eq. (*) holds because $\lambda_n\leq 1$ and therefore $\mathbb{E}[\langle z,(T+\lambda_n)^{-1}Z\rangle^2] \leq \mathbb{E}[\langle z,(T+I)^{-1}Z\rangle^2] = \sigma^2$, and the last equality holds provided that $\kappa < \frac{1}{4}(1-1/r)$. This verifies the Lyapnov condition, which implies

$$\frac{1}{\overline{\sigma}(z)\sqrt{n}}\sum_{i=1}^n \varepsilon_i\overline{g}(z,Z_i) \overset{d}{\to} \mathcal{N}(0,1), \tag{25}$$

where

$$\overline{\sigma}(z) = \sigma_\varepsilon^2\mathbb{E}[g(z,Z)^2] = \sigma_\varepsilon^2\mathbb{E}[\langle z,(T+\lambda_n I)^{-1}Z\rangle^2]$$

Using Cauchy-Schwarz inequality, it holds almost surely that

$$\begin{aligned}\left|\frac{1}{n}\sum_{i=1}^n g(z,Z_i)^2 - \overline{g}(z,Z_i)^2\right| &\leq \left|\frac{2}{n}\sum_{i=1}^n[g(z,Z_i)-\overline{g}(z,Z_i)]\overline{g}(z,Z_i)\right|\\ &\leq 2\sqrt{\frac{1}{n}\sum_{i=1}^n[g(z,Z_i)-\overline{g}(z,Z_i)]^2}\sqrt{\frac{1}{n}\sum_{i=1}\overline{g}(z,Z_i)^2}.\end{aligned}$$

Lemma A.2 then asserts that the first term on the right-hand side of the above inequality converges in probability to zero, and the second term converges in probability to $\sqrt{\mathbb{E}[g(z,Z_i)^2]}$. Therefore,

$$\widetilde{\sigma}_n(z)^2 \overset{p}{\to} \overline{\sigma}(z)^2. \tag{26}$$

Next, using Chebyshev's inequality, for any $\delta>0$, conditioned on $\{Z_i\}$ it holds with probability $1-\delta$ that

$$\left|\frac{1}{\sqrt{n}}\sum_{i=1}^n\varepsilon_i[g(z,Z_i)-\overline{g}(z,Z_i)]\right| \leq \frac{\sigma_\varepsilon^2}{\delta^2}\sqrt{\frac{1}{n}\sum_{i=1}^n[g(z,Z_i)-\overline{g}(z,Z_i)]^2}$$

Because $\frac{1}{n}\sum_{i=1}^{n}[g(z,Z_i)-\overline{g}(z,Z_i)]^2 \xrightarrow{p} 0$, for any $\epsilon, u>0$ there exists $N\in\mathbb{N}$ such that for any $n\geq N$, $\frac{1}{n}\sum_{i=1}^{n}[g(z,Z_i)-\overline{g}(z,Z_i)]^2\leq\epsilon$ with probability $1-u$. Now for any $\epsilon',\delta'>0$, pick $\delta=\delta'/2$, $u=\delta'/2$ and $\epsilon=(\epsilon')^2\delta^4/\sigma_\varepsilon^4>0$, there exists $N\in\mathbb{N}$ such that for any $n\geq N$,

$$\Pr\left[\left|\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\varepsilon_i[g(z,Z_i)-\overline{g}(z,Z_i)]\right|>\epsilon'\right]\leq\delta',$$

where the probability is on both $\{\varepsilon_i\}$ and $\{Z_i\}$, and the above inequality holds by the union bound. This implies that

$$\frac{1}{\overline{\sigma}(z)\sqrt{n}}\sum_{i=1}^{n}\varepsilon_i[g(z,Z_i)-\overline{g}(z,Z_i)] \;\xrightarrow{p}\; 0. \tag{27}$$

Combining Eqs. (25,26,27), the proof of Lemma A.3 is complete with the Slutsky's theorem. □

A.2. Proof of Theorem 4.1. To simplify notations, in the proofs of Theorems 4.1 and 4.2 we shall drop all iso superscripts as this is the only set of estimators, confidence intervals and their related quantities to be analyzed in the proofs. With $Y_i=\langle X_i,\beta_0\rangle+\varepsilon_i=\langle Z_i,\theta_0\rangle+\varepsilon_i$, it holds that

$$\widehat{\theta}_n=(T_n+\lambda_n)^{-1}T_n\theta_0+\frac{1}{n}\sum_{i=1}^{n}\varepsilon_i(T_n+\lambda_n)^{-1}Z_i.$$

Subsequently,

$$\widehat{\theta}_n-\theta_0=\underbrace{-\lambda_n(T_n+\lambda_n)^{-1}\theta_0}_{=:b_n}+\underbrace{\frac{1}{n}\sum_{i=1}^{n}\varepsilon_i(T_n+\lambda_n)^{-1}Z_i}_{=:v_n}. \tag{28}$$

For any test function $x\in\operatorname{supp}(P_X)$, let $z=L_{K^{1/2}}x=\sum_{j=1}^{n}\widehat{\nu}_j\widehat{\psi}_j$, where $\widehat{\nu}_j=\langle z,\widehat{\psi}_j\rangle$. Let also $\widehat{\vartheta}_j=\langle\theta_0,\widehat{\psi}_j\rangle$. Slightly abusing notations by denoting $b_n(z):=\langle b_n,z\rangle=\int_I b_n(t)z(t)\mathrm{d}t$, we have that

$$\big|b_n(z)\big|=\lambda_n\left|\left\langle(T_n+\lambda_n)^{-1}\sum_{j=1}^{n}\widehat{\vartheta}_j\widehat{\psi}_j,\sum_{j=1}^{n}\widehat{\nu}_j\widehat{\psi}_j\right\rangle\right|=\lambda_n\left|\sum_{j=1}^{n}\frac{\widehat{\nu}_j\widehat{\vartheta}_j}{\widehat{s}_j+\lambda_n}\right| \tag{29}$$

$$\leq\lambda_n\sqrt{\sum_{j=1}^{n}\widehat{\vartheta}_j^2}\sqrt{\sum_{j=1}^{n}\frac{\widehat{\nu}_j^2}{(\widehat{s}_j+\lambda_n)^2}} \tag{30}$$

$$\leq\sqrt{\lambda_n}\sqrt{\sum_{j=1}^{n}\widehat{\vartheta}_j^2}\sqrt{\sum_{j=1}^{n}\frac{\widehat{\nu}_j^2}{\widehat{s}_j+\lambda_n}} \tag{31}$$

$$=\sqrt{\lambda_n}\cdot\sqrt{\langle z,(T_n+\lambda_n)^{-1}z\rangle}=\frac{\sqrt{\lambda_n n}}{\sigma_\varepsilon}\widehat{\sigma}_n(z). \tag{32}$$

In the above derivations, Eq. (29) holds because $\{\widehat{\psi}_j\}_{j=1}^{n}$ is orthonormal with respect to $L_2(I)$ and $T_n=\sum_{j=1}^{n}\widehat{s}_j\widehat{\psi}_j\otimes\widehat{\psi}_j$; Eq. (30) holds by the Cauchy-Schwarz inequality; Eq. (31) holds by noting that $\lambda_n/(\sqrt{\widehat{s}_j+\lambda_n})\leq\sqrt{\lambda_n}$ for all j, and furthermore

$$\langle z,(T_n+\lambda_n)^{-1}z\rangle=\left\langle\sum_{j=1}^{n}\widehat{\nu}_j\widehat{\psi}_j,\sum_{j=1}^{n}\frac{\widehat{\nu}_j}{\widehat{s}_j+\lambda_n}\widehat{\psi}_j\right\rangle=\sum_{j=1}^{n}\frac{\widehat{\nu}_j^2}{\widehat{s}_j+\lambda_n}.$$

With $\lambda_n n \to 0$ as $n \to \infty$, Eq. (32) implies that

$$b_n(z)/\widehat{\sigma}_n(z) \;\to\; 0 \quad a.s.$$

We next analyze the term involving v_n on an arbitrary test function $x \in \mathrm{supp}(P_X)$ with $z = L_{K^{1/2}}x = \sum_{j=1}^n \widehat{\nu}_j \widehat{\psi}_j$, where $\widehat{\nu}_j = \langle z, \widehat{\psi}_j\rangle$. Let $v_n(z) = \langle v_n, z\rangle = \int_I v_n(t)z(t)\mathrm{d}t$. We have that

$$v_n(z) = \frac{1}{n}\sum_{i=1}^n \varepsilon_i \langle z, (T_n+\lambda_n)^{-1} Z_i\rangle.$$

Note that $v_n(z)$ is *pivotal*, meaning that it does not depend on the unknown $\beta_0 \in \Theta_2(\mathcal{H})$ or $\theta_0 = L_{K^{1/2}}\beta_0$. Note also that $\{\varepsilon_i\}_{i=1}^n$ are conditionally centered when $\{Z_i\}_{i=1}^n$ are fixed. Therefore, if $\{\varepsilon_i\}_{i=1}^n$ are Normally distributed, the distribution of $v_n(z)$ is an exact Normal distribution with mean zero and variance equal to

$$\frac{\sigma_\varepsilon^2}{n^2}\sum_{i=1}^n \langle z, (T_n+\lambda_n)^{-1}Z_i\rangle^2 = \frac{\sigma_\varepsilon^2}{n}\langle z, (T_n+\lambda_n)^{-1}T_n(T_n+\lambda_n)^{-1}z\rangle = \frac{\widetilde{\sigma}_n(z)^2}{n}$$

because $T_n = \frac{1}{n}\sum_{i=1}^n Z_i \otimes Z_i$. If $\{\varepsilon_i\}_{i=1}^n$ are not Normally distributed, Lemma A.3 asserts that under imposed conditions,

$$\frac{1}{\widetilde{\sigma}_n(z)\sqrt{n}}\sum_{i=1}^n \varepsilon_i\langle z, (T_n+\lambda_n)^{-1}Z_i\rangle \;\xrightarrow{d}\; \mathcal{N}(0,1),$$

which is to be proved.

A.3. Proof of Theorem 4.2. For any $x \in \mathrm{supp}(P_X)$ and $z = L_{K^{1/2}}x$, the definition of $\varphi(x)$ yields

$$n\big|\varphi(x)\big|^2 = 4\tau_{\alpha/2}^2\sigma_\varepsilon^2\langle z, (T_n+\lambda_n)^{-1}z\rangle.$$

Because $|\varphi(x)|$ is pivotal, we may safely drop the $\sup_{\beta_0\in\Theta_2(\mathcal{H})}$ notation and instead calculate

$$\begin{aligned} n\mathbb{E}_{P_X}\left[|\varphi(X)|^2\right] &= n\mathbb{E}_{P_Z}[\langle Z, (T_n+\lambda_n)^{-1}Z\rangle^2] = \sigma_\varepsilon^2 \mathrm{tr}((T_n+\lambda_n)^{-1/2}T(T_n+\lambda_n^{-1/2})) \\ &= \sigma_\varepsilon^2(1+o_P(1))\mathrm{tr}((T+\lambda_n)^{-1/2}T(T+\lambda_n)^{-1/2}), \end{aligned}$$

where the last identity holds by invoking Lemma A.1. Additionally, invoking Assumption (A1) and the fact that $\lambda_n \gtrsim n^{-1-\delta}$ for any $\delta > 0$, it holds that

$$\mathrm{tr}((T+\lambda_n)^{-1/2}T(T+\lambda_n^{-1/2})) = \sum_{j=1}^\infty \frac{s_j}{s_j+\lambda_n} \lesssim \int_1^\infty \frac{j^{-2r}}{j^{-2r}+\lambda_n}\mathrm{d}j \lesssim n^{1/2r+\delta},$$

which is to be proved.

A.4. Proof of Theorem 4.3. In this lower bound we work with $\gamma_0 = 0$ and the definition of honest confidence intervals that

$$\Pr_{\beta_0}[\langle x, \beta_0\rangle \in \varphi(x)] \geq 1-\alpha, \quad \forall \beta_0 \in \Theta_2(\mathcal{H}),\; x \in \mathrm{supp}(P_X).$$

The definition in Eq. (2) is asymptotic in nature, and would imply the above property for a slightly smaller constant significance level for sufficiently large n.

Let $\{\psi_j\}_{j=1}^{\infty}$ be an arbitrary orthonormal basis of $L_2(I)$. Consider a distribution P_X such that for every $X \sim P_X$, $\langle X, \psi_j\rangle$ are *independently distributed* random variables that satisfy

$$\Pr[\langle X, \psi_j\rangle = \sqrt{s_j}] = \Pr[\langle X, \psi_j\rangle = -\sqrt{s_j}] = \frac{1}{2}, \quad \forall j \geq 1.$$

with

$$s_j = \frac{1}{2r-1} \times j^{-2r}.$$

Let $\{\varepsilon_i\}_{i=1}^n$ be i.i.d. standard Normally distributed random variables. It is easy to verify that this problem instance satisfies Assumptions (A1), (A3), (B1), (B2), (B3), with $\sigma_\varepsilon = 1$, $M_\varepsilon = 1.6$, $c_s = 1/(2r-1)$, $C_s = 1$, $C_p = 2$, $\kappa = 0$ and $C_\nu = 1$.

Let M be an integer that grows in a particular way with n, to be specified later in this proof. Consider a model class Θ_n such that for every $\beta_0 \in \Theta_n$, either $\beta_0 \equiv 0$ or

$$\langle L_{K^{-1/2}}\beta_0, \psi_j\rangle = \begin{cases} \pm 1/\sqrt{M}, \text{ if } M < j \leq 2M; \\ 0, \qquad\qquad \text{otherwise.} \end{cases}$$

It is easy to see that $\Theta_n \subseteq \Theta_2(\mathcal{H})$. Furthermore, all $\beta_0 \in \Theta_n$ would satisfy Assumption (A2). There are a total of $2^M + 1$ hypothetical regression functionals (the one additional functional comes from $\beta_0 \equiv 0$) in Θ_n. For convenience, given any binary vector $\boldsymbol{\iota} \in \{\pm 1\}^{2M}$, define

$$\beta_{\boldsymbol{\iota}} := \sum_{j=M+1}^{2M} \frac{\boldsymbol{\iota}_j}{\sqrt{M}} L_{K^{1/2}}\psi_j.$$

Additionally, we use $\beta_o \equiv 0$ to denote the zero regression functional.

Because φ is an honest confidence interval at significance level α, it holds that

$$\Pr_{\beta_o}[0 \in \varphi(x)] \geq 1 - \alpha, \quad \forall x \in \mathrm{supp}(P_X). \tag{33}$$

Fix arbitrary $\boldsymbol{\iota} \in \{\pm 1\}^{2M}$, let $x_{\boldsymbol{\iota}} \in \mathrm{supp}(P_X)$ be such that

$$\mathrm{sgn}(\langle x_{\boldsymbol{\iota}}, \psi_j\rangle) = \boldsymbol{\iota}_j, \quad \forall j \leq 2M \tag{34}$$

and that $\langle x_{\boldsymbol{\iota}}, \psi_j\rangle = 0$ for all $j > 2M$. Note that there might be multiple such test functions satisfying the sign matching property: an arbitrary test function could be selected. Define constant a_n as

$$a_n := \langle x_{\boldsymbol{\iota}}, \beta_{\boldsymbol{\iota}}\rangle = \frac{1}{2r-1} \sum_{j=M+1}^{2M} \frac{j^{-r}}{\sqrt{M}}$$

which satisfies

$$\frac{(2M)^{-r}\sqrt{M}}{2r-1} \leq a_n \leq \frac{M^{-r}\sqrt{M}}{2r-1}. \tag{35}$$

Then, invoking again the honesty of φ, it holds that

$$\Pr_{\beta_{\boldsymbol{\iota}}}[a_n \in \varphi(x_{\boldsymbol{\iota}})] \geq 1 - \alpha. \tag{36}$$

Furthermore, for any $\beta_{\boldsymbol{\iota}} \in \Theta_2(\mathcal{H})$,

$$\mathrm{KL}(P_{\beta_o}^n \| P_{\beta_{\boldsymbol{\iota}}}^n) = n \times \mathbb{E}_{X\sim P_X}\mathbb{E}_{\beta_o}\left[\log\frac{P(Y|X,\beta_{\boldsymbol{\iota}})}{P(Y|X,\beta_o)}\right] = \frac{n}{2}\mathbb{E}_{X\sim P_X}[\langle X, \beta_{\boldsymbol{\iota}} - \beta_o\rangle^2]$$

$$= \frac{n}{2}\sum_{j=M+1}^{2M}\frac{1}{M} \times \frac{j^{-2r}}{(2r-1)^2} \leq \frac{nM^{-2r}}{2(2r-1)^2}. \tag{37}$$

Selecting

$$M \asymp n^{1/2r}$$

the right-hand side of Eq. (37) is upper bounded by 0.1, and therefore using Pinsker's inequality,

$$\Pr_{\beta_o}[a_n \in \varphi(x_{\iota})] \geq \Pr_{\beta_\iota}[a_n \in \varphi(x_{\iota})] - \sqrt{2\mathrm{KL}(P^n_{\beta_o} \| P^n_{\beta_\iota})} \geq 1/2. \tag{38}$$

Because Eq. (38) holds for any $\iota \in \{\pm 1\}^{2M}$, in conjunction with Eq. (33) we have that for each $\iota \in \{\pm 1\}^{2M}$ and the context vector x_ι defined in Eq. (34) it holds that

$$E_{\beta_o}[|\varphi(x_\iota)|^2] \geq a_n^2/2.$$

Because $\{x_\iota\}_{\iota \in \{\pm 1\}^2 M}$ uniformly explores the support of P_X that is also by definition a uniform distribution, it holds that

$$\mathbb{E}_{X \sim P_X} \mathbb{E}_{\beta_o}[|\varphi(X)|^2] \geq a_n^2/2 \gtrsim M^{1-2r} \asymp n^{-(2r-1)/2r},$$

which is to be proved.

A.5. Proof of Theorem 5.1. In the proofs of Theorems 5.1 we shall drop all ani superscripts as this is the only set of estimators, confidence intervals and their related quantities to be analyzed in these proofs. Following similar derivations that lead to Eq. (25), and accounting for the null space of $\{\widehat{\psi}_j\}_{j=1}^n$, we have for the anisotropic case that

$$\widehat{\theta}_n - \theta_0 = \underbrace{-(I - \widehat{\Pi})\theta_0}_{=:t_n} \underbrace{-(T_n + \Lambda_n)^\dagger \Lambda_n \theta_0}_{=:b_n} + \underbrace{\frac{1}{n}\sum_{i=1}^n \varepsilon_i (T_n + \Lambda_n)^\dagger Z_i}_{=:v_n}, \tag{39}$$

where $\widehat{\Pi} = \sum_{j=1}^n \widehat{\psi}_j \otimes \widehat{\psi}_j$.

Consider X_1 being an arbitrary input function in $\{(X_i, Y_i)\}_{i=1}^n$, and let $Z_1 = L_{K^{1/2}} X_1$, $b_n(Z_1) = \langle b_n, Z_1 \rangle$, $v_n(Z_1) = \langle v_n, Z_1 \rangle$. Noet that $t_n(Z_1) = 0$ because $\widehat{\Pi}(Z_1) = Z_1$. Because $v_n(z)$ is either exactly Normally distributed (if $\{\varepsilon_i\}_{i=1}^n$ are Normally distributed) PD: conditional on Z_i, or satisfies Lyapnov conditions of central limit theorem with probabilities converging to one provided that Eq. (11) is satisfied, we know that $v_n(z)$ is asymptotically Normally distributed whose variance is exactly $\frac{\sigma_\varepsilon^2}{n^2}\sum_{i=1}^n \langle z, (T_n + \Lambda_n)^\dagger Z_i \rangle^2 = \frac{\sigma_\varepsilon^2}{n}\langle z, (T_n + \Lambda_n)^\dagger T_n (T_n + \Lambda_n)^\dagger z \rangle = \widehat{\sigma}_n^{\mathsf{ani}}(z)^2$. This shows

$$\frac{v_n(z)}{\widehat{\sigma}_n^{\mathsf{ani}}(z)} \xrightarrow{d} \mathcal{N}(0, 1). \tag{40}$$

We next analyze the bias term. Using the spectral decomposition of both T_n and Λ_n, $b_n(z)$ can be expressed as

$$b_n(z) = \left\langle (T_n + \Lambda_n)^\dagger \Lambda_n \sum_{j=1}^n \widehat{\vartheta}_j \widehat{\psi}_j, \sum_{j=1}^n \widehat{\nu}_j \widehat{\psi}_j \right\rangle = \sum_{j=1}^n \frac{\lambda_{nj} \widehat{\nu}_j \widehat{\vartheta}_j}{\widehat{s}_j + \lambda_{nj}},$$

where $\widehat{\vartheta}_j = \langle \theta_0, \widehat{\psi}_j \rangle$ and $\widehat{\nu}_j = \langle z, \widehat{\psi}_j \rangle$. Using Cauchy-Schwarz inequality and the fact that $\sum_{j=1}^n \widehat{\vartheta}_j^2 \leq 1$ because $\|\theta_0\|_2^2 = \int_I \theta_0(t)^2 \mathrm{d}t \leq 1$, it holds that

$$b_n(z)^2 \leq \sum_{j=1}^n \frac{\lambda_{nj}^2 \widehat{\nu}_j^2}{(\widehat{s}_j + \lambda_{nj})^2}.$$

Using the choice that $\lambda_{nj} = \sqrt{\widehat{s}_j \lambda_n}$, the above inequality simplifies to

$$b_n(z)^2 \le \lambda_n \sum_{j=1}^{n} \frac{\widehat{\nu}_j^2 (\lambda_n \wedge \widehat{s}_j)}{(\widehat{s}_j + (\lambda_n \wedge \sqrt{\widehat{s}_j \lambda_n}))^2} \le \lambda_n \sum_{j=1}^{n} \frac{\widehat{\nu}_j^2 \widehat{s}_j}{(\widehat{s}_j + \sqrt{\widehat{s}_j \lambda_n})^2}. \tag{41}$$

On the other hand, expanding $\widehat{\sigma}_n^2(z)$ in the spectrum of $\{\widehat{\psi}_j\}_{j=1}^n$ we obtain

$$\widehat{\sigma}_n^2(z) = \frac{\sigma_\varepsilon^2}{n} \langle z, (T_n + \Lambda_n)^\dagger T_n (T_n + \Lambda_n)^\dagger z \rangle = \frac{\sigma_\varepsilon^2}{n} \sum_{j=1}^{n} \frac{\widehat{\nu}_j^2 \widehat{s}_j}{(\widehat{s}_j + \sqrt{\widehat{s}_j \lambda_n})^2}. \tag{42}$$

Comparing Eqs. (41,42) and noting that $\lambda_n n \to 0$ as $n \to \infty$, we obtain

$$\frac{|b_n(z)|}{\widehat{\sigma}_n(z)} \le \sigma_\varepsilon^{-1} \sqrt{\lambda_n n} \to 0 \quad a.s. \tag{43}$$

Eqs. (39,43) together yield Theorem 5.1, by Slutsky's theorem.

A.6. Proof of Theorem 5.2. Let $Z_i = L_{K^{1/2}} X_i$. It holds that

$$\begin{aligned}
\frac{1}{n}\sum_{i=1}^{n} \big|\varphi(X_i)\big|^2 &= \frac{4\tau_\alpha^2}{n^2} \sum_{i=1}^{n} \widehat{\sigma}_n^{\mathsf{ani}}(z_i)^2 = \frac{4\sigma_\varepsilon^2 \tau_\alpha^2}{n^2} \sum_{i=1}^{n} \langle Z_i, (T_n + \Lambda_n)^\dagger T_n (T_n + \Lambda_n)^\dagger Z_i \rangle \\
&= \frac{4\sigma_\varepsilon^2 \tau_\alpha^2}{n} \mathrm{tr}(((T_n + \Lambda_n)^\dagger T_n)^2) = \frac{4\sigma_\varepsilon^2 \tau_\alpha^2}{n} \sum_{j=1}^{n} \frac{\widehat{s}_j^2}{(\widehat{s}_j + \lambda_{nj})^2} \\
&\overset{(a)}{\le} \frac{4\sigma_\varepsilon^2 \tau_\alpha^2}{n} \sum_{j=1}^{n} \frac{\widehat{s}_j}{(\sqrt{\widehat{s}_j} + \sqrt{\lambda_n})^2} \le \frac{4\sigma_\varepsilon^2 \tau_\alpha^2}{n} \sum_{j=1}^{n} \frac{\widehat{s}_j}{\widehat{s}_j + \lambda_n} \\
&\lesssim \frac{1}{n} \int_1^\infty \frac{j^{-2r}}{j^{-2r} + \lambda_n} \mathrm{d}j \lesssim n^{-\frac{2r-1}{2r} + \delta}
\end{aligned}$$

for any $\delta > 0$, where Eq. (a) holds because $\lambda_{nj} = \sqrt{\widehat{s}_j \lambda_n}$, and the last inequality holds by the condition imposed on $\widehat{\{s_j\}}$ and the asymptotic scaling that $\lambda_n \gtrsim n^{-1-\delta}$ for any $\delta > 0$.

YW: please check again. this derivation should be correct and no longer need to do $\lambda_n \wedge \sqrt{\widehat{s}_j \lambda_n}$.

A.7. Proof of Lemma 6.1. Express both T and T_n on the eigenfunctions $\{\psi_j\}_{j=1}^n$ and obtain their infinite-dimensional matrix representations A and A_n, where $A_{ij} = \langle \psi_i, T\psi_j \rangle$ and $[A_n]_{ij} = \langle \psi_i, T_n \psi_j \rangle$ for all $i, j \ge 1$. More specifically,

$$A_{ij} = \mathbb{E}[\langle Z, \psi_i \rangle \langle Z, \psi_j \rangle] = \mathbf{1}\{i = j\} s_i.$$

On the other hand, $[A_n]_{ij}$ is the sample average of a sum of n i.i.d. random variables and is potentially non-zero for $i \ne j$:

$$[A_n]_{ij} = \frac{1}{n} \sum_{\ell=1}^{n} \langle Z_\ell, \psi_i \rangle \langle Z_\ell, \psi_j \rangle.$$

Now, let $C = (2C_s/c_s)^{1/(2r)} > 0$ be a constant such that, for every $j \in \mathbb{N}$, $s_{Cj} \le C_s (Cj)^{-2r} \le c_s j^{-2r}/2 \le s_j/2$ thanks to Assumption (B1). Let $\overline{J} = \lceil CJ(n) \rceil$. By Assumption (B2) and the Cauchy-Schwarz inequality, it holds for every $i, j \ge 1$ that

$$\mathbb{E}[\langle Z, \psi_i \rangle^2 \langle Z, \psi_j \rangle^2] \le \sqrt{\mathbb{E}[\langle Z, \psi_i \rangle^4]} \sqrt{\mathbb{E}[\langle Z, \psi_j \rangle^4]} \le C_p \mathbb{E}[\langle Z, \psi_i \rangle^2] \mathbb{E}[\langle Z, \psi_j \rangle^2] = C_p s_i s_j.$$

By Assumption (B3), it holds for every $i, j \geq 1$ that

$$|\langle Z, \psi_i \rangle \langle Z, \psi_j \rangle| \leq C_\nu^2 \sqrt{s_j s_k} n^{2\kappa}$$

almost surely. Subsequently, using Bernstein's inequality, for every $i, j \geq 1$, with probability $1 - n^{-c}/(3i^2 j^2)$ it holds that

(44)
$$\big|[A_n]_{ij} - A_{ij}\big| \leq C_1 \left(\sqrt{\frac{C_p s_i s_j \log(nij)}{n}} + \frac{C_\nu \sqrt{s_i s_j} n^{2\kappa} \log(nij)}{n} \right) \lesssim \frac{\sqrt{s_i s_j} \log(nij)}{\sqrt{n}},$$

where $C_1 < +\infty$ is a numerical constant depending only on c, and the last inequality holds provided that $\kappa < 1/4$. Using the union bound (i.e. Bonferroni's inequality) and noting that $(\sum_{j=1}^\infty 1/j^2)^2 = \pi^4/36 < 3$, with probability $1 - n^{-c}$ Eq. (44) holds for all $i, j \geq 1$.

Let $e_1, e_2, \cdots$ be the standard basis vectors. We will prove that, for every $j \in [\overline{J}]$, there exists w satisfying $\|w\|_2 = (\sum_{j=1}^\infty w_j^2)^{1/2} = O(j \log^3 n/\sqrt{n}) = o(1)$, $w_j = 0$ and $|w_k| \lesssim \Delta_\psi(j,k)$ for all $k \in \mathbb{N}\backslash\{j\}$ such that

$$A_n(e_j + w) = s_j(e_j + w), \tag{45}$$

and furthermore the convergence in the $o(1)$ term to zero as $n \to \infty$ is *uniform* over $j \in [\overline{J}]$. This shows that the self-adjoint operator $T_n = \sum_{k,\ell=1}^n [A_n]_{k\ell} \psi_k \otimes \psi_\ell$ admits a pair of eigenvalue and eigenfunction $(\widehat{s}, \widehat{\psi})$ that satisfies $\widehat{s} = s_j(1 + O(j \log^3 n/\sqrt{n})) = s_j(1 + o(1))$, $\|\widehat{\psi} - \psi_j\|_2 = O(j \log^3 n/\sqrt{n}) = o(1)$ and $|w_k| \lesssim \Delta_\psi(j,k)$ for all $k \neq j$. Furthermore:

1. These eigenfunctions of T_n are unique for different j, because an $1 - o(1)$ mass is concentrated on e_j; therefore, ϖ is an injection;
2. Because $\widehat{s}_j \geq (1 + o(1)) c_s j^{-2r}$ and $\widehat{s}_j \leq (1 + o(1)) C_s j^{-2r}$, it holds that $j/C \leq \varpi(j) \leq Cj$;
3. All eigenvalue and eigenfunction pairs $(\widehat{s}, \widehat{\psi})$ for $j \in [\overline{J}]$ will cover the top $J(n)$ eigenvalues and eigenfunctions of T_n simply from the sufficiently large number of eigenvalues; this shows that ϖ is a surjection on $[J(n)]$.

Let $H = A_n - A$ and consider $w_j = 0$ which is simply a Gauge choice. Because $Ae_j = s_j e_j$, Eq. (45) can be reformulated as

$$(A - s_j I + H)w = -He_j.$$

Let $\overline{H}$ be the submatrix (also infinite-dimensional) formed by removing the jth row and jth column of H. Let h be the jth column of H, removing H_{jj}. Let $D = \mathrm{diag}(s_1 - s_j, \cdots, s_{j-1} - s_j, s_{j+1} - s_j, \cdots)$ be a diagonal matrix of the same shape of $\overline{H}$. Let $\overline{w} = (w_1, \cdots, w_{j-1}, w_{j+1}, \cdots)$. Then

$$\overline{w} = -(D + \overline{H})^{-1} h = -D^{-1/2}(I + \widetilde{H})^{-1} D^{-1/2} h,$$

where $\widetilde{H} = D^{-1/2} \overline{H} D^{-1/2}$.

For any $i, k \neq j$, Eq. (44) implies

$$\big|\widetilde{H}_{ik}\big| \lesssim \frac{1}{\sqrt{n}} \frac{i^{-r} k^{-r} \log(nik)}{\sqrt{|(s_i - s_j)(s_k - s_j)|}} \leq \frac{1}{\sqrt{n}} z_i z_k \quad \text{where } z_k = \frac{k^{-r} \sqrt{\log n} \log(3k)}{\sqrt{|s_k - s_j|}},$$

and the $\lesssim$ inequality hides only constants and is uniform over all i, k. In a similar way, $D^{-1/2} h$ can also be upper bounded component-wise as

$$\big|[D^{-1/2} h]_k\big| \lesssim \frac{1}{\sqrt{n}} \frac{j^{-r} k^{-r} \log(njk)}{\sqrt{|s_k - s_j|}} \leq \frac{j^{-r} \log(nj)}{\sqrt{n}} z. \tag{46}$$

Discuss the following three cases:

1. The case of $k \le j/C$. In this case, by Assumption (B1), $|s_k - s_j| \ge c_s k^{-2r} - C_s(Ck)^{-2r} \ge 0.5c_s k^{-2r}$, and therefore $|z_k|/\sqrt{\log n}\log(3k) \le \sqrt{2/c_s} = O(1)$;
2. The case of $k \ge Cj$. In this case, by Assumption (B1), $|s_k - s_j| \ge c_s j^{-2r} - C_s(Cj)^{-2r} \ge 0.5c_s j^{-2r}$, and therefore $|z_k|/\sqrt{\log n}\log(3k) \le \sqrt{2/c_s} j^r/k^r = O(j^r k^{-r})$.
3. The case of $j/C \le k \le Cj$. In this case, by Assumption (D1), $|s_k - s_j| \ge |k-j| \cdot c_g(Cj)^{-2r-1}$, and therefore
$$|z_k|/\sqrt{\log n}\log(3k) \le \sqrt{C} j^{r+1/2}k^{-r}/\sqrt{c_g|k-j|} \le C^{r+1}\sqrt{c_g k/|j-k|} = O(\sqrt{k/|j-k|}).$$

Because $\widetilde{H} \le zz^\top/\sqrt{n}$, summarizing the above cases we obtain

$$\|\widetilde{H}\|_{\mathrm{op}} \lesssim \frac{1}{\sqrt{n}}\|z\|_2^2 = \frac{1}{\sqrt{n}}\sum_{k\ne j}|z_k|^2$$
$$\lesssim \frac{1}{\sqrt{n}}\left(j\log n\log^2(3j) + j^{2r}\int_j^\infty \frac{\log n\log^2(3t)\mathrm{d}t}{t^{2r}} + Cj\log n\log^2(3Cj+3)\int_1^{Cj}\frac{\mathrm{d}t}{t}\right)$$
$$\lesssim \frac{j\ln n\ln^2(j+1))}{\sqrt{n}} \lesssim \frac{j\log^3 n}{\sqrt{n}}. \tag{47}$$

With $\overline{J} \asymp n^q$ for $q \in (0,1/2)$ (Because $J(n) \asymp n^q$ and $\overline{J} = CJ$, where C is a constant), the right-hand side of the above inequality is $o(1)$ and therefore we have $\|\widetilde{H}\|_{\mathrm{op}} = o(1)$ as $n \to \infty$. We can then do the following Taylor expansion using $(I+\widetilde{H})^{-1} = \sum_{\ell=0}^\infty(-1)^\ell\widetilde{H}^\ell$, for every $k \ne j$:

$$|w_k| \le \sum_{\ell=0}^\infty \left|e_k^\top D^{-1/2}\widetilde{H}^\ell D^{-1/2}h\right| \overset{(a)}{\le} \sum_{\ell=0}^\infty \frac{j^{-r}\log(nj)}{\sqrt{n}}\left|e_k^\top D^{-1/2}\left(\frac{1}{\sqrt{n}}zz^\top\right)^\ell z\right|$$
$$\lesssim \sum_{\ell=0}^\infty \frac{j^{-r}\log(nj)}{\sqrt{n}}\left(\frac{j\log^3 n}{\sqrt{n}}\right)^\ell \left|e_k^\top D^{-1/2}z\right| \lesssim \frac{j^{-r}\log(nj)}{\sqrt{n}}\left|e_k^\top D^{-1/2}z\right|$$
$$= \frac{j^{-r}\log(nj)}{\sqrt{|s_k - s_j|n}}|z_k|. \tag{48}$$

Here, Eq. (a) holds by plugging in the expression of $D^{-1/2}h$ in Eq. (46). Invoking the case discussions earlier. For $k \le j/C$, the right-hand side of Eq. (48) is upper bounded by

$$\frac{j^{-r}\log(nj)}{\sqrt{|s_k - s_j|n}} \times \sqrt{\log n}\log(3k) \times O(1) = O\left(\frac{j^{-r}k^r\log^2 n\log(3j)\log(3k)}{\sqrt{n}}\right);$$

for $k \ge Cj$, the right-hand side of Eq. (48) is upper bounded by

$$\frac{j^{-r}\log(nj)}{\sqrt{|s_k - s_j|n}} \times \sqrt{\log n}\log(3k) \times O(j^r k^{-r}) = O\left(\frac{j^r k^{-r}\log^2 n\log(3j)\log(3k)}{\sqrt{n}}\right);$$

for $j/C \le k \le Cj$, the right-hand side of Eq. (48) is upper bounded by

$$\frac{j^{-r}\log(nj)}{\sqrt{|s_k - s_j|n}} \times O\left(\sqrt{\frac{k\log n\log^2(3k)}{|j-k|}}\right) = \frac{j^{-r}\log^2 n}{\sqrt{n}} \times O\left(\frac{j^{r+1/2}}{\sqrt{|k-j|}}\right) \times O\left(\sqrt{\frac{k\log n\log^2(3j)}{|j-k|}}\right)$$
$$= O\left(\frac{\sqrt{jk}\log^2 n\log(3j)\log(3k)}{|j-k|\sqrt{n}}\right).$$

Combining all of the above cases we complete the proof of upper bounds on $|\langle \widehat{\psi}_j, \psi_k\rangle|$ in Lemma 6.1.

Furthermore,

$$\|w\|_2 \le \frac{j^{-r}\log(nj)\sqrt{\log n}}{\sqrt{n}} \left(\int_1^j \frac{\log^3(3t)\mathrm{d}t}{t^{-2r}} + j^{2r}\int_j^{\overline{J}} \frac{t^{-2r}\log^2(3t)}{j^{-2r}}\mathrm{d}t \right.$$

$$\left. + Cj\cdot \int_1^{Cj} \frac{1}{tj^{-2r-1}} \frac{\log^2(3t)\mathrm{d}t}{t} \right)^{1/2}$$

$$\lesssim \frac{j^{-r}\log n \log^2(3Cj+3)}{\sqrt{n}} \left(j^{2r+1}+j^{2r+1}+j^{2r+2}\right)^{1/2} \lesssim \frac{j\log^3 n}{\sqrt{n}} \le \frac{\overline{J}\sqrt{\log n}}{\sqrt{n}}. \tag{49}$$

Because $\overline{J} \asymp n^q$ for some $q < 1/2$, the right-hand side of Eq. (49) is $o(1)$ uniformly for all $j \le \overline{J}$. This completes the proof.

Finally, with the additional assumption that $q < 1/4$, the right-hand side of Eq. (49) is further upper bounded by $o(1/\overline{J}) = o(1/J(n))$. Additionally, in this case we have for every $j \le \overline{J}$ that $\widehat{s}_j - \widehat{s}_{j+1} = s_j - s_{j+1} + o(s_j/\overline{J}) = s_j - s_{j+1} + o(j^{-2r}/\overline{J}) = (1+o(1))(s_j - s_{j+1})$ because $s_j - s_{j+1} \gtrsim j^{-2r-1}$. This shows that there is a one-to-one match between (s_j, ψ_j) and $(\widehat{s}_j, \widehat{\psi}_j)$ for all $j \le \overline{J}$.

A.8. Proof of Lemma 6.2. Fix arbitrary $z \in \mathrm{supp}(P_Z^n)$, $\ell \le J$ and $\ell/C_J \le j \le C_J\ell$ such that $\varpi(j) = \ell$. Recall that for $a, b > 0$, $(a+b)^M = \sum_{m=0}^M \binom{M}{m} a^m b^{M-m}$ from the binomial expansion property. Therefore, for $M \in \{2, 4\}$,

$$\left|\langle z, \widehat{\psi}_\ell\rangle^M - \langle z, \psi_j\rangle^M\right|$$

$$\le \langle z, \psi_j\rangle^M(1 - \langle \widehat{\psi}_\ell, \psi_j\rangle^M) + 6\sum_{m=0}^{M-1} \left|\langle z, \psi_j\rangle\right|^m \left(\sum_{k\neq j}\left|\langle z, \psi_k\rangle\langle \widehat{\psi}_\ell, \psi_k\rangle\right|\right)^{M-m}$$

$$\le 2\langle z, \psi_j\rangle^M \sum_{k\neq j}\langle \widehat{\psi}_\ell, \psi_k\rangle^2 + 6\sum_{m=0}^{M-1} \left|\langle z, \psi_j\rangle\right|^m \left(\sum_{k\neq j}\left|\langle z, \psi_k\rangle\langle \widehat{\psi}_\ell, \psi_k\rangle\right|\right)^{M-m} \tag{50}$$

$$\le 2L_J^M \left[\langle z, \psi_j\rangle^M \sum_{k\neq j}\Delta_\psi(j,k)^2 + 3\sum_{m=0}^{M-1}\left|\langle z, \psi_j\rangle\right|^m \left(\sum_{k\neq j}\left|\langle z, \psi_k\rangle\right|\Delta_\psi(j,k)\right)^{M-m}\right]. \tag{51}$$

Here, Eq. (50) holds because $1 - \langle \widehat{\psi}_\ell, \psi_j\rangle^2 = \sum_{k\neq j}\langle \widehat{\psi}_\ell, \psi_k\rangle^2$ from the Pythagorean theorem, and $1 - \langle \widehat{\psi}_\ell, \psi_j\rangle^4 = (1 + \langle \widehat{\psi}_\ell, \psi_j\rangle^2)(1 - \langle \widehat{\psi}_\ell, \psi_j\rangle^2) \le 2(1 - \langle \widehat{\psi}_\ell, \psi_j\rangle^2)$. By Assumption (B3), $|\langle z, \psi_k\rangle| \le C_\nu \sqrt{s_k} n^\kappa \lesssim k^{-r}n^\kappa$ almost surely. Subsequently, the right-hand side of Eq. (51) is upper bounded almost surely, up to constants depending only on q and other parameters in assumptions, by

$$n^{M\kappa}\left[j^{-rM}\sum_{k\neq j}\Delta_\psi(j,k)^2 + \sum_{m=0}^{M-1} j^{-rm}\left(\sum_{k\neq j}k^{-r}\Delta_\psi(j,k)\right)^{M-m}\right]. \tag{52}$$

Using expressions of $\Delta_\psi(\cdot,\cdot)$, we have that

$$\sum_{k\neq j}\Delta_\psi(j,k)^2 \lesssim \frac{\log^2 n\log^2(3j)}{n}\times\Bigg(\frac{1}{j^{2r}}\int_1^{C_J j}k^{2r}\log^2(3k)\mathrm{d}k$$

$$+j^{2r}\int_{C_J j}^{\infty}\frac{\log^2(3k)\mathrm{d}k}{k^{2r}}+3C_J j^2\log^2(3C_J j)\cdot\int_1^{C_j j}\frac{\mathrm{d}t}{t^2}\Bigg)$$

$$\lesssim \frac{j^2\log^6 n}{n}, \tag{53}$$

and

$$\sum_{k\neq j}k^{-r}\Delta_\psi(j,k) \lesssim \frac{\log n\log(3j)}{\sqrt{n}}\times\Bigg(\frac{1}{j^{r}}\int_1^{C_J j}k^{-r}\cdot k^{r}\log(3k)\mathrm{d}k$$

$$+j^{r}\int_{C_J j}^{\infty}k^{-r}\cdot\frac{\log(3k)\mathrm{d}k}{k^{r}}+\frac{(3C_J)^{1/2}j\log(3C_J j)}{(j/C_j)^r}\cdot\int_1^{C_j j}\frac{\mathrm{d}t}{t}\Bigg)$$

$$\lesssim \frac{j^{1-r}\log^4 n}{\sqrt{n}}. \tag{54}$$

Subsequently,

$$\sum_{m=0}^{M-1}j^{-rm}\left(\sum_{k\neq j}k^{-r}\Delta_\psi(j,k)\right)^{M-m} \lesssim \sum_{m=0}^{M-1}j^{-rm}\left(\frac{j^{1-r}\log^4 n}{\sqrt{n}}\right)^{M-m}$$

$$= \sum_{m=0}^{M-1}j^{-rM}\left(\frac{j\log^4 n}{\sqrt{n}}\right)^{M-m} \lesssim \frac{J^{1-rM}\log^4 n}{\sqrt{n}}, \tag{55}$$

where the last inequality holds because $j\log^4 n/\sqrt{n}\leq J\log^4 n/\sqrt{n}=O(n^{1/2-q}\log^4 n)\to 0$ and as $n\to\infty$, and therefore the $m=M-1$ term dominates. Combining Eqs. (52,53,55) and noting that $j\lesssim n^q$ for some $q<1/2$, we complete the proof of the first statement in Lemma 6.2.

Finally, because the right-hand side of Eq. (51) is a low-degree polynomial of $\{\langle z,\psi_k\rangle\}_{k=1}^\infty$, to prove the second inequality involving expectations, simply replace Assumption (B3) with Assumption (B2) and remove the $n^{M\kappa}$ factor. More specifically, for the second inequality, set $z=Z$ in Eq. (50) and take expectations. By Assumption (B2), $\mathbb{E}\,|\langle Z,\psi_k\rangle|^p\lesssim s_k^{p/2}$ for $p=2,4$. Combining these moment bounds with Cauchy Schwarz over $\sum_{k\neq j}|\langle Z,\psi_k\rangle|\,|\langle\widehat{\psi}_\ell,\psi_k\rangle|$ and the bound established in Eq. (52) we obtain the second inequality involving expectations, which is the same bound as the first inequality with the factor $n^{M\kappa}$ removed.

A.9. Technical lemmas for results in Secs. 6 and 7..

LEMMA A.4. *Let $J=J(n)$ and $\underline{J}=J(n)/C_J$. Conditioned on results in Lemma 6.1, it holds that*

$$\mathbb{E}_{Z\sim P_Z^n}[\|\widehat{\Pi}_J^{\perp}Z\|_2^2]\lesssim J^{1-2r}\left(1+\frac{J\log^4 n}{\sqrt{n}}\right),$$

where $\widehat{\Pi}_J^{\perp}=I-\sum_{j=1}^J\widehat{\psi}_j\otimes\widehat{\psi}_j$.

PROOF OF LEMMA A.4. Fix arbitrary $z \in L_2(I)$ and, without loss of generality, assume $\|z\|_2 = 1$. Throughout this proof, all expectations are taken with respect to Z. The Pythagorean theorem asserts that

$$\|z\|_2^2 = \|\widehat{\Pi}_J z\|_2^2 + \|\widehat{\Pi}_J^\perp z\|_2^2 = \|\Pi_{\underline{J}} z\|_2^2 + \|\Pi_{\underline{J}}^\perp z\|_2^2, \tag{56}$$

where $\widehat{\Pi}_J = \sum_{j=1}^J \widehat{\psi}_j \otimes \widehat{\psi}_j$, $\Pi_{\underline{J}} = \sum_{j=1}^{\underline{J}} \psi_j \otimes \psi_j$ and $\widehat{\Pi}_J^\perp = I - \widehat{\Pi}_J$, $\Pi_{\underline{J}}^\perp = I - \Pi_{\underline{J}}$. Using Lemmas 6.1 and 6.2, it holds that

$$\begin{aligned}
\mathbb{E}[\|\widehat{\Pi}_J Z\|_2^2] &= \sum_{j=1}^J \mathbb{E}[\langle Z, \widehat{\psi}_j\rangle^2] \geq \sum_{j=1}^{\underline{J}} \mathbb{E}[\langle Z, \widehat{\psi}_{\varpi(j)}\rangle^2] \\
&\geq \left(\sum_{j=1}^{\underline{J}} \mathbb{E}[\langle Z, \psi_j\rangle^2]\right) - O\left(\frac{J^{2-2r}\log^4 n}{\sqrt{n}}\right) \\
&= \mathbb{E}[\|\Pi_{\underline{J}} Z\|_2^2] - O\left(\frac{J^{2-2r}\log^4 n}{\sqrt{n}}\right)
\end{aligned} \tag{57}$$

Combining Eqs. (56,57) and noting that $\mathbb{E}[\|\Pi_{\underline{J}}^\perp Z\|_2^2 \lesssim \int_{\underline{J}}^\infty j^{-2r}\mathrm{d}j \lesssim \overline{J}^{1-2r}$, we complete the proof of Lemma A.4, in the following manner:

$$\mathbb{E}[\|\widehat{\Pi}_J^\perp Z\|_2^2] = \mathbb{E}[\|\Pi_{\underline{J}} Z\|_2^2] + \mathbb{E}[\Pi_{\underline{J}}^\perp Z\|_2^2] - \mathbb{E}[\|\widehat{\Pi}_J Z\|_2^2],$$

where $\mathbb{E}[\|\Pi_{\underline{J}} Z\|_2^2]$ cancels out the same term in the expression of Eq. (57). ☐

The following lemma holds for *any* choices of Λ_J.

LEMMA A.5. *Let $J = J(n)$ and suppose $q \in (0, 1/2)$ and $\kappa < (1-q)/2$. Conditioned on the results in Lemma A.4, it holds that*

$$\sup_{x \in \mathrm{supp}(P_X^n)} \frac{\sum_{i=1}^n |g(z, Z_i)|^3}{(\sum_{i=1}^n g(z, Z_i)^2)^{3/2}} \lesssim \frac{\sqrt{J}n^\kappa}{\sqrt{n}}\left(1 + \frac{\sqrt{J}\log^2 n}{\sqrt[4]{n}}\right) \to 0 \quad \textit{as } n \to \infty,$$

where $z = L_{K^{1/2}}x$, $Z_i = L_{K^{1/2}}X_i$ and $g(z, Z) = \langle z, (T_J + \Lambda_J)^\dagger Z\rangle$.

PROOF OF LEMMA A.5. Let $\varsigma^2 = \frac{1}{n}\sum_{i=1}^n g(z, Z_i)^2 = \langle z, (T_J + \Lambda_J)^\dagger T_n (T_J + \Lambda_J)^\dagger z\rangle = \langle z, (T_J + \Lambda_J)^\dagger T_J (T_J + \Lambda_J)^\dagger z\rangle = \|T_J^{1/2}(T_J + \Lambda_J)^\dagger z\|_2^2$. Note that here in the second equality, the middle operator T_n is switched to T_J, which is valid because $(T_J + \Lambda_J)^\dagger$ is fully within the span of $\{\widehat{\psi}_j\}_{j=1}^J$ by the definition of Moore-penrose pseudo inverses on operators. Using Hölder's inequality, we have that

$$\begin{aligned}
\frac{\sum_{i=1}^n |g(z, Z_i)|^3}{(\sum_{i=1}^n g(z, Z_i)^2)^{3/2}} &\leq \frac{\max_{1\leq i\leq n}|g(z, Z_i)|}{\sqrt{n}\varsigma} = \max_{1\leq i\leq n} \frac{|\langle z, (T_J + \Lambda_J)^\dagger Z_i\rangle|}{\sqrt{n}\|T_J^{1/2}(T_J + \Lambda_J)^\dagger z\|_2} \\
&\leq \max_{1\leq i\leq n} \frac{\|T_J^{\dagger/2} Z_i\|_2}{\sqrt{n}},
\end{aligned} \tag{58}$$

where Eq. (58) holds by using the Cauchy-Schwarz inequality on $\langle z, (T_J + \Lambda_J)^\dagger Z_i\rangle$ to cancel the demoninator, and $T_J^{\dagger/2} = (T_J^\dagger)^{1/2} = \sum_{j=1}^J (\widehat{s}_j)^{-1/2}\widehat{\psi}_j \otimes \widehat{\psi}_j$.

By the results of Lemma A.4, for any $\ell \le J$, there exists $j \le \overline{J} = C_J J$ such that $\varpi(j) = \ell$. We denote $\varpi^{-1}(\ell) = j$ in this case. Using also Lemma 6.2, it holds almost surely that

$$
\begin{aligned}
\|T_J^{\dagger/2} Z_i\|_2^2 &= \sum_{\ell=1}^{J} \frac{\langle Z_i, \widehat{\psi}_\ell\rangle^2}{\widehat{s}_\ell} \le \sum_{\ell=1}^{J} \frac{2\langle Z_i, \widehat{\psi}_\ell\rangle^2}{s_{\varpi^{-1}(\ell)}} \\
&\le \sum_{\ell=1}^{J} \frac{2}{s_{\varpi^{-1}(\ell)}} \left[\langle Z_i, \psi_{\varpi^{-1}(\ell)}\rangle^2 + O\left(\frac{n^{2\kappa} J^{1-2r} \log^4 n}{\sqrt{n}}\right)\right] \\
&\lesssim \sum_{j=1}^{\overline{J}} O(n^{2\kappa}) \times \left[1 + \frac{1}{s_j} \cdot \frac{J^{1-2r} \log^4 n}{\sqrt{n}}\right] \\
&\lesssim J n^{2\kappa} + J^2 n^{2\kappa} \log^4 n / \sqrt{n}, \qquad (59)
\end{aligned}
$$

where the last inequality holds because $\sum_{j=1}^{\overline{J}} \frac{1}{s_j} \lesssim \int_1^{\overline{J}} j^{2r} \mathrm{d}j \lesssim \overline{J}^{1+2r} \lesssim J^{1+2r}$. Combining Eqs. (58,59) we complete the proof of Lemma A.5. □

LEMMA A.6. *There exists a constant n_0 depending only on parameters on assumptions, such that for all $n \ge n_0$, conditioned on results in Lemma 6.2, the following inequality holds for all $\epsilon > 0$:*

$$
\Pr_{Z\sim P_Z}\left[\sigma(Z)^2 \ge \Omega(\epsilon) \times \sum_{j=1}^{J} \frac{s_j}{s_j + \lambda_n} - O\left(\frac{J^{2-2r}\log^4 n}{\epsilon\sqrt{n}}\right)\right] \ge 1 - O(\epsilon),
$$

where $\sigma(Z)^2 = \langle Z, (T_J + \Lambda_J)^\dagger T_J (T_J + \Lambda_J)^\dagger Z\rangle$, and $\widehat{s}_j + \lambda_n$ is an upper bound on the diagonal entries of Λ_J. Constants in all $O(\cdot), \Omega(\cdot)$ notations depend only on parameters in assumptions.

REMARK A.1. If Assumptions (B1) and (C2) hold with $q < 1/2$, then all terms in big-O notations in Lemma A.6 can be safely dropped, at a cost of perhaps larger n_0 that still depends only on parameters in assumptions.

PROOF OF LEMMA A.6. Let $J = J(n)$. It holds that

$$
\sigma(Z)^2 = \sum_{j=1}^{J} \frac{\langle Z, \widehat{\psi}_j\rangle^2 \widehat{s}_j}{(\widehat{s}_j + \lambda_{nj})^2} = (1 + o(1)) \sum_{j=1}^{J} \frac{\langle Z, \widehat{\psi}_j\rangle^2 s_{\varpi^{-1}(j)}}{(s_{\varpi^{-1}(j)} + \lambda_{nj})^2}.
$$

Let $b_j := s_{\varpi^{-1}(j)}/(s_{\varpi^{-1}(j)} + \lambda_{nj})$. Define

$$
\overline{\sigma}(Z)^2 := \sum_{j=1}^{J} b_j \langle Z, \psi_j\rangle^2 + \Delta(Z)
$$

where $\Delta(Z)$ is a random variable defined as

$$
\Delta(Z) := \sum_{j=1}^{J} b_j (\langle Z, \widehat{\psi}_j\rangle^2 - \langle Z, \psi_j\rangle^2).
$$

Conditioned on the results in Lemma 6.2, it holds that

$$\mathbb{E}_{Z\sim P_Z}[|\Delta(Z)|] \lesssim \sum_{j=1}^{J} b_j \frac{J^{1-2r}\log^4 n}{\sqrt{n}} \leq \frac{J^{2-2r}\log^4 n}{\sqrt{n}}.$$

Subsequently, using Markov's inequality, for sufficiently large n it holds for every $\epsilon > 0$ that

$$\Pr_{Z\sim P_Z}\left[\sigma(Z)^2 \geq 0.9\sum_{j=1}^{J} b_j\langle Z,\psi_j\rangle^2 - O(\epsilon^{-1}) \times \frac{J^{2-2r}\log^4 n}{\sqrt{n}}\right] \geq 1-\epsilon. \tag{60}$$

Note that $\lambda_{nj}(z) \leq \sqrt{\widehat{s}_j\lambda_n}$ in both $\widehat{\sigma}_n^{\mathsf{anitr}}$ and $\widehat{\sigma}_n^{\mathsf{adptr}}$ (for the latter, replace λ_n with μ_n). This implies

$$b_j = \frac{s_{\varpi(j)}}{s_{\varpi(j)} + \lambda_{nj}} \geq \frac{s_{\varpi(j)}}{s_{\varpi(j)} + \sqrt{\widehat{s}_j\lambda_n}} \geq \Omega(1) \times \frac{s_j}{s_j + \lambda_n},$$

where the last inequality holds by invoking the first two properties of Lemma 6.1 and the AM-GM inequality that $\sqrt{\widehat{s}_j\lambda_n} \leq (\widehat{s}_j + \lambda_n)/2$. Let $a_j := s_j/(s_j+\lambda_n)$. Then the above lower bound on b_j implies, from Eq. (60), that

$$\Pr_{Z\sim P_Z}\left[\sigma(Z)^2 \geq \Omega(1) \times \sum_{j=1}^{J} a_j\langle Z,\psi_j\rangle^2 - O(\epsilon^{-1}) \times \frac{J^{2-2r}\log^4 n}{\sqrt{n}}\right] \geq 1-\epsilon. \tag{61}$$

Additionally, Assumption (C3) yields for every $\epsilon > 0$ that

$$\Pr_{Z\sim P_Z}\left[\sum_{j=1}^{J} a_j\langle Z_j,\psi_j\rangle^2 \geq \epsilon\sum_{j=1}^{J} a_j s_j\right] \geq 1 - C_\epsilon\epsilon. \tag{62}$$

Combining Eqs. (61,62) and using the Bonferroni's correction we complete the proof of Lemma A.6. □

A.10. Proof of Theorem 6.1. Let $J = J(n)$, $T_J = \sum_{j=1}^{J}\widehat{s}_j\widehat{\psi}_j \otimes \widehat{\psi}_j$ and $\Lambda_J = \sum_{j=1}^{J}\lambda_{nj}\widehat{\psi}_j\otimes\widehat{\psi}_j$. Decompose the estimation error of $\widehat{\theta}_n^{\mathsf{anitr}} - \theta_0$ as

$$\widehat{\theta}_n^{\mathsf{anitr}} - \theta_0 = \underbrace{-\widehat{\Pi}_J^{\perp}\theta_0}_{=:t_n}\underbrace{-(T_J+\Lambda_J)^{\dagger}\Lambda_J\theta_0}_{=:b_n} + \underbrace{\frac{1}{n}\sum_{i=1}^{n}\varepsilon_i(T_J+\Lambda_J)^{\dagger}Z_i}_{=:v_n}, \tag{63}$$

where $\widehat{\Pi}_J^{\perp} = I - \sum_{j=1}^{J}\widehat{\psi}_j\otimes\widehat{\psi}_j$.

The rest of the proof is carried out fixing an arbitrary $x \in \mathrm{supp}(P_X)$ and $z = L_{K^{1/2}}x$. The truncation error is precisely $t_n(z) = \langle z, t_n\rangle = -\langle z, \widehat{\Pi}_J^{\perp}\theta_0\rangle = -\langle\widehat{\Pi}_J^{\perp}PD : z, \theta_0\rangle$, because $\widehat{\Pi}_J^{\perp}$ is a self-disjoint, projection operator.

For the bias term $b_n(z) = \langle z, b_n\rangle$, define $\widehat{\nu}_j := \langle z, \widehat{\psi}_j\rangle$ and $\widehat{\vartheta}_j := \langle\theta_0, \widehat{\psi}_j\rangle$ for all $j \leq J$. Recall also the definition that $\lambda_{nj} = \sqrt{\widehat{s}_j\lambda_n}$ Then

$$|b_n(z)|^2 = \left|\sum_{j=1}^{J}\frac{\lambda_{nj}\widehat{\vartheta}_j\widehat{\nu}_j}{\widehat{s}_j+\lambda_{nj}}\right|^2 \leq \sum_{j=1}^{J}\frac{\lambda_{nj}^2\widehat{\nu}_j^2}{(\widehat{s}_j+\lambda_{nj})^2} = \lambda_n\sum_{j=1}^{J}\frac{\widehat{\nu}_j^2\widehat{s}_j}{(\widehat{s}_j+\sqrt{\widehat{s}_j\lambda_n})^2} \tag{64}$$

$$= \lambda_n\langle z, (T_J+\Lambda_J)^{\dagger}T_J(T_J+\Lambda_J)^{\dagger}z\rangle. \tag{65}$$

Here, the second inequality in Eq. (64) holds by using the Cauchy-Schwarz inequality and noting that $\sum_{j=1}^J \widehat{\vartheta}_j^2 \leq \|\theta_0\|_2^2 \leq 1$. Because $\lambda_n n \to 0$ as $n \to \infty$, Eq. (65) implies

$$\sup_{x\in\mathrm{supp}(P_X^n)} b_n(z)/\widehat{\sigma}_n^{\mathsf{anitr}}(z) \;\to\; 0 \quad a.s. \tag{66}$$

as $n \to \infty$, where $z = L_{K^{1/2}}x$.

For the variance term v_n, let $v_n(z) = \langle v_n, z\rangle = \frac{1}{n}\sum_{i=1}^n \varepsilon_i\langle z, (T_J + \Lambda_J)^\dagger Z_i\rangle$. Using Lemma A.5 and the finite-sample Berry-Esseen theorem for independently but not identically distributed random variables, there exists a constant $C < +\infty$ depending only on parameters of Assumptions, such that

$$\sup_{x\in\mathrm{supp}(P_X^n)} \|\widehat{\sigma}_n^{\mathsf{anitr}}(z)^{-1}v_n(z) - u\|_{\mathrm{KS}} \;\leq\; C\left(n^{-c} + \frac{\sqrt{J}n^\kappa}{\sqrt{n}}\left(1 + \frac{\sqrt{J}\log^2 n}{\sqrt[4]{n}}\right)\right), \tag{67}$$

where $z = L_{K^{1/2}}x$, $u \sim \mathcal{N}(0,1)$ and $\|P - Q\|_{\mathrm{KS}} := \sup_{t\in\mathbb{R}}|F_P(t) - F_Q(t)|$ is the Kolmogorov-Smirnov (KS) distance between two uni-variate distributions P and Q, with F_P and F_Q being their CDFs respectively. The first term in Eq. (67) accounts for the failure probability in Lemma 6.1. Under the conditions imposed, the right-hand side of Eq. (67) converges to zero as $n \to \infty$. Combining Eqs. (66,67) we complete the proof of the first statement in Theorem 6.1.

To prove the strengthened second statement in Theorem 6.1 under the additional conditions, first note that, using the Markov inequality, the second property of Lemma A.4 implies for every $\epsilon > 0$ that

$$\Pr_{Z\sim P_Z^n}\left[\|\widehat{\Pi}_J^\perp Z\|_2 > \frac{CJ^{\frac{1}{2}-r}}{\epsilon}\left(1 + \frac{\sqrt{J}\log^2 n}{\sqrt[4]{n}}\right)\right] \;\leq\; \epsilon/2, \tag{68}$$

where $C < +\infty$ is a constant depending only on constants in the assumptions. On the other hand, for sufficiently large n, it holds for all $\epsilon > 0$ that

$$\Pr_{Z\sim P_Z^n}\left[\widehat{\sigma}_n^{\mathsf{anitr}}(Z) > \frac{C'_\epsilon\sigma_\varepsilon}{\sqrt{n}}\left(\sum_{j=1}^J \frac{s_j}{s_j + \lambda_n} - C''_\epsilon \times \frac{J^{2-2r}\log^4 n}{\sqrt{n}}\right)^{1/2}\right] \geq 1 - \epsilon/2, \tag{69}$$

where $0 < C'_\epsilon, C''_\epsilon < +\infty$ only depends on ϵ and other assumption parameters. With $J \asymp n^q$ and $\frac{1}{2r} < q < \frac{1}{2}$, it holds that $\sum_{j=1}^J \frac{s_j}{s_j+\lambda_n} \asymp \lambda_n^{-1/(2r)}$ and $\frac{J^{2-2r}\log^4 n}{\sqrt{n}} \lesssim n^{(2-2r)q-1/2} = o(\lambda_n^{-1/(2r)})$ provided that $\lambda_n n \to 0$ and $\lambda_n n^{1+\delta} \to \infty$ for all $\delta > 0$. Comparing Eqs. (68,69) and using the union bound, with probability $1 - \epsilon$ it holds that

$$\frac{\|\widehat{\Pi}_J^\perp Z\|}{\widehat{\sigma}_n^{\mathsf{anitr}}(Z)} \lesssim \sqrt{n}\lambda_n^{1/4r}J^{\frac{1}{2}-r} \to 0,$$

provided that $q > 1/2r$. This completes the proof.

A.11. Proof of Theorem 6.2. Let $J = J(n) \asymp n^q$ for some $q \in (0, 1/2)$. Without explicitly mentioned, all expectations and probabilities in this proof are with respect to the randomness in $Z \sim P_Z^n$. Invoking Lemma A.4, it holds that $\mathbb{E}[\|\widehat{\Pi}_J^\perp Z\|_2^2] \lesssim J^{1-2r} \lesssim n^{-q(2r-1)}$. This explains the second term in Theorem 6.2.

Now let us focusing only on the $\sigma(Z)^2 = \langle Z, (T_J + \Lambda_J)^\dagger T_J(T_J + \Lambda_J)^\dagger Z\rangle$ term in the definition of $\widehat{\sigma}_n^{\mathsf{anitr}}(Z)$. Using spectral representations, it holds that

$$\mathbb{E}[\sigma(Z)^2] = \sum_{j=1}^J \frac{\widehat{s}_j\mathbb{E}[\langle Z, \widehat{\psi}_j\rangle^2]}{(\widehat{s}_j + \lambda_{nj})^2} = \sum_{j=1}^J \frac{\mathbb{E}[\langle Z, \widehat{\psi}_j\rangle^2]}{\widehat{s}_j + \lambda_n}. \tag{70}$$

Applying Lemmas 6.1 and 6.2, with probability $1-O(n^{-c})$ it holds for sufficiently large n that

$$\begin{aligned}\sum_{j=1}^{J}\frac{\mathbb{E}[\langle Z,\widehat{\psi}_j\rangle^2]}{\widehat{s}_j+\lambda_n} &\leq \sum_{j=1}^{J}\frac{2}{s_{\varpi^{-1}(j)}+\lambda_n}\left[s_{\varpi^{-1}(j)}+O\left(\frac{J^{1-2r}\log^4 n}{\sqrt{n}}\right)\right]\\ &\lesssim \int_1^{\infty}\frac{j^{-2r}\mathrm{d}j}{j^{-2r}+\lambda_n}+O\left(\frac{J^{1-2r}\log^4 n}{\sqrt{n}}\right)\times\int_1^{C_J j}\frac{\mathrm{d}j}{j^{-2r}}\\ &\lesssim \lambda_n^{-1/2r}+O\left(\frac{J^2\log^4 n}{\sqrt{n}}\right). \qquad (71)\end{aligned}$$

With $J\asymp n^q$ and $\lambda_n=\omega(n^{-1-\delta})$ for any $\delta>0$, we complete the proof of Theorem 6.2.

A.12. Proof of Theorem 7.1. The analysis of the truncation and variance terms are identical to the proof in Theorem 6.1. In the rest of this proof, we shall therefore focus on the bias term primarily, and additional constraints leading to the strengthened statement dropping truncation terms.

Let $J=J(n)$ and $\overline{J}=C_J J$. Fix arbitrary $x\in\mathrm{supp}(P_X^n)$ and let $z=L_{K^{1/2}}x$. Let $b_n(z)=\langle z,b_n\rangle$ be the bias term that is represented as

$$b_n(z)=-\sum_{j=1}^{J}\frac{\lambda_{nj}\widehat{\vartheta}_j\widehat{\nu}_j}{\widehat{s}_j+\lambda_{nj}},$$

where $\widehat{\vartheta}_j=\langle\theta_0,\widehat{\psi}_j\rangle$ and $\widehat{\nu}_j=\langle z,\widehat{\psi}_j\rangle$. Using Hölder's inequality and Assumption (D1), it holds almost surely that

$$\big|b_n(z)\big|\leq\left(\sum_{j=1}^{J}|\vartheta_j|\right)\times\max_{1\leq j\leq J}\frac{\lambda_{nj}|\widehat{\nu}_j|}{\widehat{s}_j+\lambda_{nj}}. \qquad (72)$$

Recall the definition that $\theta_k=\langle\theta_0,\psi_k\rangle$ for every $k\geq 1$. Because $q<1/2$, the map ϖ in Lemma 6.1 can be chosen such that it is a surjection on $[J]$, meaning that for every $j\in[J]$, there exists a unique $\ell\leq C_J j$ such that $\varpi(\ell)=j$. For notational simplicity let ϱ be the inverse function of ϖ on $[J]$, so that $\varpi(\varrho(j))=j$ for all $j\leq J$. Conditioned on the results in Lemma 6.1, it holds for every $j\leq J$ that

$$\begin{aligned}\big|\widehat{\vartheta}_j\big| &= \big|\langle\theta_0,\widehat{\psi}_{\varrho(j)}\rangle\big|\leq\big|\langle\theta_0,\psi_{\varrho(j)}\rangle\big|+\sum_{k\neq\varrho(j)}\big|\langle\theta_0,\psi_k\rangle\langle\widehat{\theta}_\ell,\psi_k\rangle\big|=\big|\theta_j\big|+\sum_{k\neq\varrho(j)}\big|\theta_k\langle\widehat{\psi}_\ell,\psi_k\rangle\big|\\ &\leq\big|\theta_{\varrho(j)}\big|+\sqrt{\sum_{k\neq\varrho(j)}\langle\widehat{\psi}_\ell,\psi_k\rangle^2}\leq\big|\theta_{\varrho(j)}\big|+L_J\sqrt{\sum_{k\neq\varrho(j)}\Delta_\psi(\varrho(j),k)^2} \qquad (73)\\ &\leq\big|\theta_{\varrho(j)}\big|+O\left(\frac{j\log^3 n}{\sqrt{n}}\right) \qquad (74)\end{aligned}$$

where the first inequality in Eq. (73) holds by using the Cauchy-Schwarz inequality and noting that $\theta_0\in\Theta_1(\mathcal{H})\subseteq\Theta_2(\mathcal{H})$ implies $\sum_{j=1}^{\infty}\theta_j^2<+\infty$, and the second inequality in Eq. (73) holds by applying Lemma 6.1; the last inequality holds by plugging in the expressions of $\Delta_\psi(\varrho(j),k)$, noting that $\varrho(j)\leq C_J j=O(j)$ and following the analysis that leads to Eq. (53) in the proof of Lemma 6.2. With $J(n)\asymp n^q$ for $q\in(0,1/2)$, both ϱ and ϖ are injections and subsequently Eqs. (72,74) imply that

$$\big|b_n(z)\big|\leq C\times\max_{1\leq j\leq J}\frac{\lambda_{nj}|\widehat{\nu}_j|}{\widehat{s}_j+\lambda_{nj}} \qquad (75)$$

for some $C > 0$ depending only on assumption parameters.

For every $j \le J$, conduct a case analysis:

1. If $\widehat{\nu}_j^2 \ge 2\widehat{s}_j$, then $\lambda_{nj} = \widehat{s}_j\sqrt{\mu_n}/|\widehat{\nu}_j|$ and therefore
$$\frac{\lambda_{nj}|\widehat{\nu}_j|}{\widehat{s}_j + \lambda_{nj}} \le \frac{\widehat{s}_j\sqrt{\mu_n}}{\widehat{s}_j + \lambda_{nj}} \le \sqrt{\mu_n};$$
2. If $\widehat{\nu}_j^2 < 2\widehat{s}_j$, then $\lambda_{nj} = \sqrt{\widehat{s}_j\mu_n}$ and therefore
$$\frac{\lambda_{nj}|\widehat{\nu}_j|}{\widehat{s}_j + \lambda_{nj}} = \frac{\sqrt{\widehat{s}_j\mu_n}\sqrt{2\widehat{s}_j}}{\widehat{s}_j + \sqrt{\widehat{s}_j\mu_n}} = \frac{\sqrt{2\mu_n}}{1 + \sqrt{\mu_n/\widehat{s}_j}} \le \sqrt{2\mu_n};$$

Combining the above inequalities with Eq. (75), we have
$$\big|b_n(z)\big| \le C \times \sqrt{\mu_n},$$
which proves the upper bound on $|b_n(z)|$.

Next, we show that under additional conditions ω_n can be set to zero. Note that, with the choice of $\lambda_{nj}(z)$, it always holds that $\lambda_{nj}(z) \le 2\sqrt{\widehat{s}_j\mu_n}$. Subsequently,
$$\widehat{\sigma}_n^{\mathsf{adptr}}(z)^2 \ge \frac{\sigma_\varepsilon^2}{n}\sum_{j=1}^{J}\frac{\langle z, \widehat{\psi}_j\rangle^2\widehat{s}_j}{(\widehat{s}_j + 2\sqrt{\widehat{s}_j\mu_n})^2}.$$

Invoking Lemma A.6, under the conditions that $\frac{1}{2r+1} < q < \frac{1}{2}$, for any $\epsilon > 0$ it holds with probability at least $1 - \epsilon/2$ for all sufficiently large n that
$$\widehat{\sigma}_n^{\mathsf{adptr}}(z)^2 \gtrsim n^{-1}\mu_n^{-1/2r}.$$

Combining this with Eq. (68) in the proof of Theorem 6.1, with probability $1 - \epsilon$ it holds for sufficiently large n that
$$\frac{\|\widehat{\Pi}_J^{\perp}Z\|_2 + |b_n(z)|}{\widehat{\sigma}_n^{\mathsf{adptr}}(Z)} \lesssim \sqrt{n}\mu_n^{1/4r}J^{1/2-r} + \sqrt{n}\mu_n^{1/4r}\sqrt{\mu_n} \to 0,$$
provided that $q > 2r/((2r-1)(2r+1))$.

A.13. Proof of Theorem 7.2. The $n^{-q(2r-1)+\delta}$ and n^{1-c} terms arise from the truncation errors and failure probabilities of Lemma 6.1, which enjoy the same analysis and upper bounds in the proof of Theorem 6.2. Furthermore, $\omega_n\mu_n \lesssim n^{-\frac{2r}{2r+1}+\delta}$ for any $\delta > 0$ simply from the asymptotic scalings of ω_n and μ_n. Therefore, we shall focus entirely on the $\widehat{\sigma}_n^{\mathsf{adptr}}(z)^2$ term in the rest of this proof.

Let $J = J(n)$. It holds that
$$\widehat{\sigma}_n^{\mathsf{adptr}}(z)^2 \le \frac{\sigma_\varepsilon^2}{n}\sum_{j=1}^{J}\frac{\widehat{\nu}_j(z)^2\widehat{s}_j}{(\widehat{s}_j + \lambda_{nj}(z))^2} \le \frac{\sigma_\varepsilon^2}{n}\sum_{j=1}^{J}\frac{\widehat{\nu}_j(z)^2}{\widehat{s}_j + \lambda_{nj}(z)}$$
$$\le \frac{\sigma_\varepsilon^2}{n}\sum_{j=1}^{J}\frac{\widehat{\nu}_j(z)^2}{\widehat{s}_j + \sqrt{\widehat{s}_j\mu_n}} + \frac{\sigma_\varepsilon^2}{n}\sum_{j=1}^{J}\frac{\widehat{\nu}_j(z)^2}{\widehat{s}_j + \widehat{s}_j\sqrt{\mu_n}/|\widehat{\nu}_j(z)|} \tag{76}$$
$$\le \underbrace{\frac{\sigma_\varepsilon^2}{n}\sum_{j=1}^{J}\frac{\widehat{\nu}_j(z)^2}{\widehat{s}_j + \sqrt{\widehat{s}_j\mu_n}}}_{=:\mathcal{A}(z)} + \underbrace{\frac{\sigma_\varepsilon^2}{n}\sum_{j=1}^{J}\frac{|\widehat{\nu}_j(z)|^3}{\widehat{s}_j(|\widehat{\nu}_j(z)| + \sqrt{\mu_n})}}_{=:\mathcal{B}(z)}. \tag{77}$$

where Eq. (76) by splitting the two case of $\widehat{\nu}_j^2(z) \geq 2\widehat{s}_j$, $\widehat{\nu}_j^2(z) < 2\widehat{s}_j$ and including both of them in the upper bound.

For term $\mathcal{A}$, conditioned on the results of Lemma 6.1 with ϱ being the inverse function of injection ϖ on $[J]$, we upper bound its expectation as

$$\mathbb{E}_{Z\sim P_Z^n}[\mathcal{A}(Z)] \leq \sum_{j=1}^{J} \frac{\mathbb{E}[\widehat{\nu}_j(Z)^2]}{\widehat{s}_j + \mu_n} \leq \sum_{j=1}^{J} \frac{2\mathbb{E}[\widehat{\nu}_j(Z)^2]}{s_{\varrho(j)} + \mu_n}. \tag{78}$$

for sufficiently large n.

For term $\mathcal{B}$, we upper bound its expectation as

$$\mathbb{E}_{Z\sim P_Z^n}[\mathcal{B}(Z)] \leq \sum_{j=1}^{J} \frac{\sqrt{\mathbb{E}\left[\widehat{\nu}_j(Z)^4\right]}}{\widehat{s}_j} \sqrt{\mathbb{E}\left[\frac{\widehat{\nu}_j(Z)^2}{(|\widehat{\nu}_j(Z)| + \sqrt{\mu_n})^2}\right]} \tag{79}$$

$$\leq \sum_{j=1}^{J} \frac{\sqrt{\mathbb{E}\left[\widehat{\nu}_j(Z)^4\right]}}{\widehat{s}_j} \sqrt{\mathbb{E}\left[\frac{\widehat{\nu}_j(Z)^2}{\widehat{\nu}_j(Z)^2 + \mu_n}\right]} \tag{80}$$

$$\leq \sum_{j=1}^{J} \frac{\sqrt{\mathbb{E}\left[\widehat{\nu}_j(Z)^4\right]}}{\widehat{s}_j} \sqrt{\frac{\mathbb{E}[\widehat{\nu}_j(Z)^2]}{\mathbb{E}[\widehat{\nu}_j(Z)^2] + \mu_n}} \tag{81}$$

$$\leq \sum_{j=1}^{J} \frac{2\sqrt{\mathbb{E}\left[\widehat{\nu}_j(Z)^4\right]}}{s_{\varrho(j)}} \sqrt{\frac{\mathbb{E}[\widehat{\nu}_j(Z)^2]}{\mathbb{E}[\widehat{\nu}_j(Z)^2] + \mu_n}} \tag{82}$$

Here, Eq. (79) holds by Cauchy-Schwarz inequality; Eq. (80) holds by the fact that $(a+b)^2 \geq a^2 + b^2$ for $ab \geq 0$; Eq. (81) holds from the Jensen's inequality and the fact that $f(x) = x/(a+x)$ is a concave function of x on $\mathbb{R}_+$ for every $a > 0$; Eq. (83) holds by applying Lemma 6.1 for sufficiently large n.

Next, conditioned on results in Lemma 6.2 and the fact that $\varpi(j) = j$ since $q < 1/4$, for every $j \leq J$ we have

$$\mathbb{E}[\widehat{\nu}_j(Z)^2] = \mathbb{E}[\langle Z, \psi_j\rangle^2] + O\left(\frac{J^{1-2r}\log^4 n}{\sqrt{n}}\right) \in [0.5s_j, 2s_j]; \tag{83}$$

$$\mathbb{E}[\widehat{\nu}_j(Z)^4] \leq \mathbb{E}[\langle Z, \psi_j\rangle^4] + O\left(\frac{J^{1-2r}\log^4 n}{\sqrt{n}}\right) \leq 2C_p s_j^2, \tag{84}$$

where the second inequality in Eq. (84) holds thanks to Assumption (B2). Because $s_J \gtrsim J^{-2r}$, with the condition that

Combining Eqs. (77), (78), (82), (83), and (84), we obtain

$$\begin{aligned}
\mathbb{E}_{Z\sim P_Z^n}\left[\widehat{\sigma}_n^{\mathsf{adptr}}(Z)\right]^2 &\leq \frac{\sigma_\varepsilon^2}{n}\left(\sum_{j=1}^{J} \frac{4s_j}{s_{\varrho(j)} + \mu_n} + \sum_{j=1}^{J} \frac{2\sqrt{C_p} s_{\varrho(j)}^{3/2}}{0.5 s_{\varrho(j)}(\sqrt{s_{\varrho(j)}} + \sqrt{\mu_n})}\right) \\
&\lesssim \frac{1}{n}\left(\int_1^\infty \frac{j^{-2r}\mathrm{d}j}{j^{-2r} + \mu_n} + \int_1^\infty \frac{j^{-r}\mathrm{d}j}{j^{-r} + \sqrt{\mu_n}}\right) \\
&\lesssim \frac{\mu_n^{-1/2r}}{n} \lesssim n^{-\frac{2r}{2r+1}+\delta},
\end{aligned}$$

for any $\delta > 0$, where the second inequality holds because $j/C_J \leq \varrho(j) \leq C_J j$ implying that $\varrho(j)$ is on the same order of j up to a numerical constant of C_J that is uniform for all j. This proves the theorem.

REFERENCES

CAI, T. T. and HALL, P. (2006). Prediction in functional linear regression. *The Annals of Statistics* **34** 2159–2179.

CAI, T. T. and LOW, M. G. (2006). Adaptive confidence balls. *The Annals of Statistics* **34** 202–228.

CAI, T. T. and YUAN, M. (2012). Minimax and adaptive prediction for functional linear regression. *Journal of the American Statistical Association* **107** 1201–1216.

CAPONNETTO, A. and DE VITO, E. (2007). Optimal rates for the regularized least-squares algorithm. *Foundations of Computational Mathematics* **7** 331–368.

CHERNOZHUKOV, V., CHETVERIKOV, D. and KATO, K. (2014). Anti-concentration and honest, adaptive confidence bands. *The Annals of Statistics* **42** 1787–1818.

FONGI, G. and GONZALEZ, M. C. (2023). Moore-Penrose inverse and partial orders on Hilbert space operators. *Linear Algebra and its Applications* **674** 1–20.

HSU, D., KAKADE, S. M. and ZHANG, T. (2014). Random design analysis of ridge regression. *Foundations of Computational Mathematics* **14** 569–600.

LIU, M., SHANG, Z. and YANG, Y. (2025). Scalable inference for nonparametric stochastic approximation in reproducing kernel Hilbert spaces. *The Annals of Statistics (accepted)*.

MINSKER, S. (2017). On some extensions of Bernstein's inequality for self-adjoint operators. *Statistics & Probability Letters* **127** 111–119.

RAMSAY, J. O. and SILVERMAN, B. W. (2005). *Functional data analysis*. Springer.

SMALE, S. and ZHOU, D.-X. (2007). Learning theory estimates via integral operators and their approximations. *Constructive Approximation* **26** 153–172.

YUAN, M. and CAI, T. T. (2010). A reproducing kernel Hilbert space approach to functional linear regression. *The Annals of Statistics* **38** 3412–3444.